

Program overview

19-May-2025 11:35

Year 2024/2025
Organization Civil Engineering and Geosciences
Education Minors Civil Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Minors CT 2024	Minors Civil Engineering 2024		
CT-MI-147-24	CT-MI-147-24 Bend and Break		
CT3170-19	Bend and Break Experimentele Recycling	5	
CT3170-19 Toets 1	Theorie	1,75	
CT3170-19 Toets 2	Practicum	3,25	
CT3260-24	Bend and Break Beton	5	
CT3260-24 Toets 1	Individual written exam	1	
CT3260-24 Toets 2	Practical work	2	
CT3260-24 Toets 3	Report practical work	2	
CT3270-15	Bend and Break: Form and Force	5	
CT3270-15 Toets 1	Theorie	1	
CT3270-15 Toets 2	Practicum	4	
CT3280-19	Bend and Break Masonry	5	
CT3280-19 Toets 1	Theory	1,5	
CT3280-19 Toets 2	Practical	3,5	
CT3295-19	Bend and Break Asphalt	5	
CT3295-19 Toets 1	Theorie	1,5	
CT3295-19 Toets 2	Practicum	3,5	
CT3602	Analysis of the Energy, Materials and Water Cycle in the Linear and Circular Economy	5	
CT3602 Toets 1	Exam	2,5	
CT3602 Toets 2	Report	2,5	
CT-MI-166-24	CT-MI-166-24 Delta Expert: Water for the Future		
CT3016	Project Minor Water Research	9	
CT3367	Introduction to Water and Climate	4	
CT3411-16	Waterworks in Practice	5	
CT3412-20	Measurements for Water	4	
CTB3360	Water System Analysis	4	
CTB3366	Rainfall in the City	4	
CTB3366 Toets 1	Practical	2,8	
CTB3366 Toets 2	Exam	1,2	
CT-MI-170-24	CT-MI-170-24 Study abroad CT		
CT-MI-174-24	CT-MI-174-24 Project Management: from Nano to Mega		
CT3101	Project Management Basics	5	
CT3102-15	Introduction to project finance & legal aspects of projects	5	
CT3102-15 Toets 1	Legal Aspects	2,5	
CT3102-15 Toets 2	Project finance	2,5	
CT3103	Integration: technical project	10	
CT3104	Fit-for-purpose projectmanagement	5	
TBM610A	Process Management and Decision-making in Project Environments	5	
CT-MI-186-24	CT-MI-186-24 Integrated Infrastructure Design (IID)		
BK7932	Introduction to Integrated Infrastructure Design	3	
BK7933	Idiosyncratic Infrastructures	3	
BK7934	Design of (infra-)structures	6	
BK7935	Environment & Infrastructures: Urban Systems + Landscape	6	
CT3200	Infrastructure Planning & Governance	3	
CT3201	Interdisciplinary & Collaborative Design Project	9	
CT-MI-188-24	CT-MI-188-24 Schakelminor CT voor BK		
CT3182	Module 5 Levenscyclus analyse	3	
CTB1210	Dynamica & Modelvorming	5	
CTB1310	Construction Mechanics 2	5	
CTB2200	Kansrekening & Statistiek	3	
CT-MI-220-24	CT-MI-220-24 African Dynamics		
CT-MI-226-24	CT-MI-226-24 Climate Change, Adaptation and Mitigation (CAM)		
CT3080	The Science of Climate Change	5	
CT3080 Toets 1	Final exam	3,5	
CT3080 Toets 2	Group assignment	1,5	
CT3081-24	Climate Change Impact and Adaptation	5	
CT3081-24 Toets 1	Exam	4,5	
CT3081-24 Toets 2	Assignment	0,5	

CT3082	Mitigation of Climate Change	5	
CT3082 Toets 1	Final exam	3,5	
CT3082 Toets 2	Group assignment	1,5	
CT3083-24	Climate Engineering	4	
CT3083-24 Toets 1	Group Assignment	2,8	
CT3083-24 Toets 2	Individual Assignment	1,2	
CT3084	Political Economy of Climate Change	5	
CT3084 Toets 1	Final exam	3,5	
CT3084 Toets 2	Group assignment	1,5	
CT3085	End-Project	6	
CT3085 Toets 1	Group report	4	
CT3085 Toets 2	Group Presentation & Discussion	2	
CT-MI-229-24			
AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
CT3601	Exploratory Research Assignment	10	
CT3601 Toets 1	Voorontwerp Q1	1	
CT3601 Toets 2	Groepsrapportage	7	
CT3601 Toets 3	Eindpresentatie	1	
CT3601 Toets 4	Reflectieverslag	1	
CT3601 Toets 5	Peer assessment	0	
CT3602	Analysis of the Energy, Materials and Water Cycle in the Linear and Circular Economy	5	
CT3602 Toets 1	Exam	2,5	
CT3602 Toets 2	Report	2,5	
CT3603-23	Solid Waste to Secondary Material Resources	5	
CT3603-23 Toets 1	Computer practicum	1	
CT3603-23 Toets 2	Tentamen	4	
CT3604	Biomass Nutrients and Energy Conversion	2	
CTB2121	Urban Water and Environmental Technology	5	
CTB2121 Toets 1	Deeltentamen Stedelijk Water	0	
CTB2121 Toets 2	Deeltentamen Milieutechniek	0	
CTB2121 Toets 3	Integraal tentamen	3	
CTB2121 Toets 4	Integrale opdracht	2	
CTB2121 Toets 5	Workshop	0	
EE3105TU	Sustainable Energy Technologies	3	
CT-MI-234-24			
CT3501	Mobiliteit en Bereikbaarheid	4	
CT3502	Het Transportsysteem	4	
CT3503	Stad en Transport	4	
CT3505-24	Traffic Flows and Management	4	
CT3505-24 Toets 1	Traffic flow modelling and management - from dynamics to control policy	2	
CT3505-24 Toets 2	Mobility management system - conceptual model	2	
CT3510	Onderzoeksproject Mobiliteit	7	
CT3520	Seminars in M&T: Onderzoeksvelden en Trends	3	
CT3521	Seminars in M&T: Case Studies	4	
Vrije minor extern project 10/15 EC			
CTBAESB3010	Free minor project	10	
CTBAESB3015	Free minor project	15	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

Minors CT 2024	
Contact for students	Please contact the minor coordinator
Director of Education	Ir. T.J. Zitman
Program Coordinator	Richard Beugeling EducationSupport-CEG@tudelft.nl

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-147-24	
Responsible Program Employee	Dr.ir. M.C.M. Bakker
Contact for students	Coordinator (responsible teacher) of each participating course
Administration by the Faculty of	CITG
Minor Title	Bend and Break
Intended for	Students from TUD, UT, TUE, Leiden (only beta studies) and Erasmus (only beta studies)
Expected prior Knowledge	Affinity with numerical and analytical analysis software packages is an advantage.
Prerequisites Minor	Finished propaedeutic year, at least the BSA, from a BSc that provides basic knowledge of physics, mechanics and materials.
Minor Exit Qualifications	After following this minor you will have: - Gained knowledge and hands-on experience with structural properties and material properties including recycled building materials. - Gained knowledge of structural modelling. - Sharpened both your analytic and testing skills. - Gained experience with accuracy of scientific models in relation to real behaviour of structures and materials. - Gained experience in group dynamics by working in multidisciplinary teams on challenging design problems.
Minor Coherence / Goal	The themes cover all the main aspects of material behaviour and structural design up to the realisation of a test structure, including the end-of life phase and recycling to new building materials.
Minor Content 1	In this minor you will work on design problems for load-bearing constructions and critical parts from road and rail infrastructure, and you will learn about specific properties of primary and recycled building materials. Example topics are recycling and reuse of reinforced concrete, constructing with glass, an architectural construction in steel or gypsum, testing of interaction between train and rails, and the behaviour of an asphalt road surface. Collaborating in a small multidisciplinary group you will come up with solutions, analyse, build and test, sometimes to collapse of your specimen. To support the design and analysis processes, workshops and lectures are given, possibly supplemented with an excursion to experience real practice. The lectures also serve to introduce the discipline so that you can place what you have learned in context. Important goals are that you get new skills and the practical experience supports your self-confidence when you go into practice as a design engineer.
Education Methods	Lectures, workshops, and practical work (supervised or independent) in computer rooms and laboratory.
Minor Assessment	Exact details will be announced by the responsible teacher during the first lecture and on Brightspace. At least 20% of the final grade must be related to individual assessment of the theory. CT3255-19 Bend en Break Rail Engineering, 5 EC, P1, lab tests and reporting (70%) and written exam (30%). CT3170-19 Bend and Break Experimental recycling, 5 EC, P1, lab test case and computer practicum reports (65%) and written exam (35%). CT3295-19 Bend en Break Asphalt, 5 EC, P1, lab tests and reporting (70%) and testing of theory (30%). CT3260-15 Bend en Break Concrete, 5 EC, P1-2, lab tests and reporting (80%) and written exam (20%). CT3280-19 Bend en Break Masonry, 5 EC, P2, lab tests and reporting (70%) and written exam (30%). CT3270-15 Bend en Break Form and Force, 5 EC, P2, lab tests and reporting (80%) and written exam (20%). A written exam is administered twice a year: first following the lectures and second in the subsequent period. A testing of theory is individual and may have the form of a written exam, an intermediate test, or an individual contribution to a report.
Maximum number of participants	maximum 50 students
Minimum number of participants	30
Minor Remarks / Schedule	Total 30 ECTS, Periods I and II

CT3170-19	Bend and Break Experimentele Recycling	5
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	6 colleges + 5 dagdelen/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Recycling of building materials in theory en practice; waste materials and recycling routes; materials separation principles and technology; quality analysis; granular material sampling and professional testing. A practicum is provided on concrete waste recycling into new concrete and a computer practicum on optimization of a complete recycling plant.	
Study Goals	Knowledge of recycling: processes, techniques and waste materials; knowledge of quality assessment and value of recycled materials; hands-on experience with concrete demolition waste processing.	
Education Method	Lectures, lab-practicum in Stevin-2 and computer practicum.	
Assessment	65% for reports on practical. works and 35% for the written exam. Minimum grade for both parts is 5.75	
Expected prior Knowledge	BSc level mechanics and materials.	
Academic Skills	Acquire knowledge and insights in material behaviour and material value; get experience with experimental research on analysis of materials.	
Literature & Study Materials	Lecture sheets and additional materials on Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Usual writing tools and calculator, but no laptop or communication devices.	
Collegerama	No	

CT3170-19 Toets 1	Theorie	1.75
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	Check CT3170-19	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check CT3170-19	
Study Goals	Check CT3170-19	
Education Method	Check CT3170-19	
Assessment	Check CT3170-19	
Expected prior Knowledge	Check CT3170-19	
Academic Skills	Check CT3170-19	
Literature & Study Materials	Check CT3170-19	
Judgement	Check CT3170-19	
Permitted Materials during Exam	Check CT3170-19	
Collegerama	No	

CT3170-19 Toets 2	Practicum	3.25
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	Check CT3170-19	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check CT3170-19	
Study Goals	Check CT3170-19	
Education Method	Check CT3170-19	
Assessment	Check CT3170-19	
Expected prior Knowledge	Check CT3170-19	
Academic Skills	Check CT3170-19	
Literature & Study Materials	Check CT3170-19	
Judgement	Check CT3170-19	
Permitted Materials during Exam	Check CT3170-19	
Collegerama	No	

CT3260-24	Bend and Break Beton	5
Responsible Instructor	Dr. F. Zhang	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	-	
Education Period	1	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Design, analysis, construction and testing of a reinforced concrete deep beam made of recycled concrete.	
Study Goals	At the end of this course you will be able to describe the real deformation and failure behaviour of structural parts, model load carrying structures, function efficiently in teams working on challenging design problems, apply analytical skills in a practical manner, analyse the accuracy of scientific models.	
Education Method	Working sessions and self study.	
Assessment	Structure, experimental result and report.	
Expected prior Knowledge	Basic mechanics and material behavior	
Academic Skills	Setting up a research question, perform experiments and evaluate results.	
Literature & Study Materials	Provided during lectures.	
Judgement	40% Report, 40% experiment and 20% theory For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator and formula sheet.	
Collegerama	No	

CT3260-24 Toets 1	Individual written exam	1
Responsible Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	-	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See CT3260-24	
Study Goals	See CT3260-24	
Education Method	Lectures	
Assessment	Written exam	
Expected prior Knowledge	See CT3260-24	
Academic Skills	See CT3260-24	
Literature & Study Materials	Materials distributed during lectures	
Judgement	See CT3260-24	
Permitted Materials during Exam	1 sided, hand written A4 with formula's	
Collegerama	No	

CT3260-24 Toets 2	Practical work	2
Responsible Instructor	N.G. Ozerkan	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	Check CT3260-24	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check CT3260-24	
Study Goals	Check CT3260-24	
Education Method	Check CT3260-24	
Assessment	Check CT3260-24	
Expected prior Knowledge	Check CT3260-24	
Academic Skills	Check CT3260-24	
Literature & Study Materials	Check CT3260-24	
Judgement	Check CT3260-24	
Permitted Materials during Exam	Check CT3260-24	
Collegerama	No	

CT3260-24 Toets 3	Report practical work	2
Responsible Instructor	N.G. Ozerkan	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	See CT3260-24	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See CT3260-24	
Study Goals	See CT3260-24	
Education Method	See CT3260-24	
Assessment	See CT3260-24	
Expected prior Knowledge	See CT3260-24	
Academic Skills	See CT3260-24	
Literature & Study Materials	See CT3260-24	
Judgement	See CT3260-24	
Permitted Materials during Exam	See CT3260-24	
Collegerama	No	

CT3270-15	Bend and Break: Form and Force	5
Responsible Instructor	Ir. H. Rozendaal	
Instructor	Prof.dr. M. Overend	
Contact Hours / Week x/x/x/x	nvt xxxxxx nvt	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Design, analysis, fabrication and testing of a special spatial structure.	
Study Goals	At the end of this course you will be able to: 1. Explain and evaluate real deformation and failure behaviour of load-bearing structures. 2. Model and analyse three-dimensional load-bearing structures digitally. 3. Function efficiently in a team when working on a challenging design problem. 4. Design and build an efficient structure for a given set of criteria.	
Education Method	Lectures, workshops, exercises and self-study.	
Computer Use	FEM software and Rhino/Grasshopper.	
Reader	Manual available via Brightspace.	
Assessment	Design presentation, testing, report, written exam.	
Expected prior Knowledge	Structural design, structural mechanics and linear algebra.	
Academic Skills	Analysing a problem and setting-up a method for solving it. Writing a report.	
Literature & Study Materials	Available via Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator	
Collegerama	No	

CT3270-15 Toets 1	Theorie	1
Responsible Instructor	Ir. H. Rozendaal	
Contact Hours / Week x/x/x/x	See CT3270-15	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Theoretical aspects of designing and calculating special spatial structures	
Study Goals	At the end of this course you will be able to: 1. Explain and evaluate real deformation and failure behaviour of load-bearing structures. 2. Model and analyse three-dimensional load-bearing structures digitally. 3. Function efficiently in a team when working on a challenging design problem. 4. Design and build an efficient structure for a given set of criteria.	
Education Method	Lectures	
Assessment	Written exam	
Expected prior Knowledge	Structural design, structural mechanics and linear algebra.	
Academic Skills	Analysing a problem and setting-up a method for solving it.	
Literature & Study Materials	Available via Brightspace.	
Judgement	Written exam	
Permitted Materials during Exam	Calculator	
Collegerama	No	

CT3270-15 Toets 2	Practicum	4
Responsible Instructor	Ir. H. Rozendaal	
Contact Hours / Week x/x/x/x	See CT3270-15	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Design, analysis, fabrication and testing of a special spatial structure.	
Study Goals	At the end of this course you will be able to: 1. Explain and evaluate real deformation and failure behaviour of load-bearing structures. 2. Model and analyse three-dimensional load-bearing structures digitally. 3. Function efficiently in a team when working on a challenging design problem. 4. Design and build an efficient structure for a given set of criteria.	
Education Method	Workshops, exercises, self-study.	
Assessment	Design presentation, testing, report.	
Expected prior Knowledge	Structural design, structural mechanics and linear algebra.	
Academic Skills	Writing a report	
Literature & Study Materials	Assailable via Brightspace	
Judgement	Design presentation, testing, report.	
Permitted Materials during Exam	Calculator	
Collegerama	No	

CT3280-19	Bend and Break Masonry	5
Responsible Instructor	Dr.ir. P.A. Korswagen Eguren	
Instructor	Prof.dr.ir. J.G. Rots	
Contact Hours / Week x/x/x/x	NA	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	An extensive overview of current masonry bricks and mortars and a glimpse into the usage of masonry in the past. Details of current masonry construction practices in the Netherlands and worldwide. Strength design of masonry using analytical models. The course also discusses masonry and earthquakes in Groningen, and gives an overview of failure, damage and strengthening strategies for masonry. Emphasis is given to the theory and practice of the construction of masonry arches, with a real-scale practicum of masonry (arch) construction and testing.	
Study Goals	After the course students will be able to: 1) compare different masonry types and structures (used in The Netherlands in typical buildings) 2) Discuss which properties of masonry are advantages or pitfalls when used as a building material in current structures. 3) explain the mechanics of simple arches and walls 4) design and build masonry arch (for experimenting in the lab) 5) employ a design guideline for assessment of masonry (structures) 6) apply good practices when building with common masonry	
Education Method	Lectures and lab practicum.	
Assessment	Summative lab practicum report and written final exam. Judgement: 70% Summative lab practicum report and 30% written final exam	
Tags	Building Engineering Structural Mechanics	
Expected prior Knowledge	Finished propedeuse and BSc knowledge of mechanics and materials.	
Academic Skills	Reading and Listening, Communication, Team Work, Organisation and Time Management.	
Literature & Study Materials	Lecture sheets, reader, self-evaluation questions, and additional background materials are made available on Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Basic calculator and drawing aids	
Collegerama	No	

CT3280-19 Toets 1	Theory	1.5
Responsible Instructor	Dr.ir. P.A. Korswagen Eguren	
Instructor	Prof.dr.ir. J.G. Rots	
Contact Hours / Week x/x/x/x	NA	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Check CT3280-19	
Study Goals	Check CT3280-19	
Education Method	Check CT3280-19	
Assessment	Check CT3280-19	
Expected prior Knowledge	Check CT3280-19	
Academic Skills	Check CT3280-19	
Literature & Study Materials	Check CT3280-19	
Judgement	Check CT3280-19	
Permitted Materials during Exam	Check CT3280-19	
Collegerama	No	

CT3280-19 Toets 2	Practical	3.5
Responsible Instructor	Dr.ir. P.A. Korswagen Eguren	
Instructor	Prof.dr.ir. J.G. Rots	
Contact Hours / Week x/x/x/x	Check CT3280-19	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Check CT3280-19	
Study Goals	Check CT3280-19	
Education Method	Check CT3280-19	
Assessment	Check CT3280-19	
Expected prior Knowledge	Check CT3280-19	
Academic Skills	Check CT3280-19	
Literature & Study Materials	Check CT3280-19	
Judgement	Check CT3280-19	
Permitted Materials during Exam	Check CT3280-19	
Collegerama	No	

CT3295-19	Bend and Break Asphalt	5
Responsible Instructor	A. Varveri	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	4-6 colleges + 2-4 dagdelen practicum	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The CT3295-15 Minor Bend and Break Asphalt course will address the following topics: Review of the role and function of asphalt concrete pavements, the most commonly used pavement type in the Netherlands. Theory of pavement design and pavement materials. Theory of multi-layer pavement analysis. Physical and mechanical properties of asphalt concrete. Mix design and preparation of asphalt samples. Stiffness measurements using 4-point-bending test and ultra sonic measurements. Utilization of computational pavement analysis software for determination of pavement response.	
Study Goals	By the end of this course, you will be able to: - prepare your own asphalt sample; - determine the density and air voids content; - translate experimental results of asphalt concrete mixtures into pavement response predictions using a computational pavement analysis software; - explain the difference in stiffness as determined by diverse measurement techniques (4-point-bending and 3-point-bending tests, and ultra sonic measurements).	
Education Method	On-campus lectures & laboratory sessions.	
Assessment	Group report and written final exam.	
Tags	Lab Research Project	
Expected prior Knowledge	Mechanics of materials, basic computer skills	
Academic Skills	- Critical and analytical thinking - Report writing - Collaboration and team-work	
Literature & Study Materials	- Course slides - Handouts	
Judgement	Final grade is based 70% on the group report and 30% on the written exam. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator	
Collegerama	No	

CT3295-19 Toets 1	Theorie	1.5
Responsible Instructor	A. Varveri	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	Check CT32915-19	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check CT32915-19	
Study Goals	Check CT32915-19	
Education Method	Check CT32915-19	
Assessment	Check CT32915-19	
Expected prior Knowledge	Check CT32915-19	
Academic Skills	Check CT32915-19	
Literature & Study Materials	Check CT32915-19	
Judgement	Check CT32915-19	
Permitted Materials during Exam	Check CT32915-19	
Collegerama	No	

CT3295-19 Toets 2	Practicum	3.5
Responsible Instructor	A. Varveri	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	Check CT32915-19	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Check CT32915-19	
Study Goals	Check CT32915-19	
Education Method	Check CT32915-19	
Assessment	Check CT32915-19	
Expected prior Knowledge	Check CT32915-19	
Academic Skills	Check CT32915-19	
Literature & Study Materials	Check CT32915-19	
Judgement	Check CT32915-19	
Permitted Materials during Exam	Check CT32915-19	
Collegerama	No	

CT3602	Analysis of the Energy, Materials and Water Cycle in the Linear and Circular Economy	5
Responsible Instructor	Prof.dr. H.M. Jonkers	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	To properly understand the transition from a linear to a circular economy, it is necessary that the current linear model in which raw materials are used and discarded is well understood. LCA and energy analysis are useful tools that have been developed to map the sustainability of different processes in society. The analysis will show that throwing away products makes it virtually impossible to set up a sustainable society. Just like in nature, we will therefore have to extract all useful elements from waste products in order to be sustainable, for example by directly reusing the entire product or recycling components. This means that we must work towards a circular economic model in which waste must be valued as a new raw material. To achieve this transition, interventions must be made to reuse the substances. A case in form of an autonomous system will be worked out in which both required amounts of water, materials and energy and environmental performance of the system will be quantified.	
Study Goals	LO1 Applying Life Cycle Assessment (LCA) techniques for quantification of environmental impact of utility structures and civil engineering works (infra- and hydraulic structures) LO2 Quantification of environmental impact of renewable materials and energies LO3 Identification and characterization of circular materials LO4 Design of low environmental impact utility constructions and civil engineering works LO5 Identify bottlenecks and find solutions for successful market implementation of innovative circular- and low environmental impact materials	
Education Method	Lectures and workshops related to the 5 defined learning objectives of the course: 1. LCA techniques, 2. Renewable materials and energies, 3. Circular materials, 4. Integral design of multi-functional low environmental impact constructions and 5. Bottlenecks for successful market implementation of circular construction materials and products.	
Assessment	Case study portfolio grade contributes 30% and the written examination 70% to the final course grade. Minimum final grade must be 5,75 in order to pass the course.	
Expected prior Knowledge	None	
Academic Skills	Students learn how to quantify the environmental impact of building materials, processes and utility buildings and civil engineering works over their entire life cycle including use of water and energy. Quantification of environmental impact is done using Life Cycle Assessment (LCA) methods.	
Literature & Study Materials	Provided scientific publications and lectures on LO1-5 subjects	
Judgement	Case study portfolio grade contributes 30% and the written examination 70% to the final course grade. Minimum final grade must be 5,75 in order to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/ .	
Permitted Materials during Exam	Dictionary, calculator	
Collegerama	No	

CT3602 Toets 1	Exam	2.5
Responsible Instructor	Prof.dr. H.M. Jonkers	
Contact Hours / Week x/x/x/x	See Course Description: CT3602	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	See Course Description: CT3602	
Study Goals	See Course Description: CT3602	
Education Method	See Course Description: CT3602	
Assessment	See Course Description: CT3602	
Expected prior Knowledge	See Course Description: CT3602	
Academic Skills	See Course Description: CT3602	
Literature & Study Materials	See Course Description: CT3602	
Judgement	See Course Description: CT3602	
Permitted Materials during Exam	See Course Description: CT3602	
Collegerama	No	

CT3602 Toets 2		Report	2.5
Responsible Instructor	Prof.dr. H.M. Jonkers		
Contact Hours / Week x/x/x/x	See Course Description: CT3602		
Education Period	1		
Start Education	1		
Exam Period	none		
Course Language	English		
Course Contents	See Course Description: CT3602		
Study Goals	See Course Description: CT3602		
Education Method	See Course Description: CT3602		
Assessment	See Course Description: CT3602		
Expected prior Knowledge	See Course Description: CT3602		
Academic Skills	See Course Description: CT3602		
Literature & Study Materials	See Course Description: CT3602		
Judgement	See Course Description: CT3602		
Permitted Materials during Exam	See Course Description: CT3602		
Collegerama	No		

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-166-24

Minor Coordinator	Dr. E. Mostert
Contact for students	For more Information: Dr. Erik Mostert Faculteit Civiele Techniek en Geowetenschappen Afdeling Watermanagement T: +31 (0)15 27 87800 E: e.mostert@tudelft.nl Dr. Ir. Matthieu de Schipper Faculteit Civiele Techniek en Geowetenschappen Afdeling Waterbouwkunde T: +31 (0)15 27 89220 E: m.a.deschipper@tudelft.nl
Program Title	Delta Expert, Water for the Future
Director of Education	Ir. T.J. Zitman
Program Coordinator	Richard Beugeling (r.e.beugeling@tudelft.nl)
In association with the Faculty of	CITG
Contact for Students	Pascal de Smidt Kamer 2.77, tel. 015 27 85742 Marian Roodenburg Kamer 2.77.1, tel. 015 27 85742 Anneloes Klapwijk Kamer 2.79, tel: 015 27 85742 Inlooppreekuur voor vragen korter dan 5 min De studieadviseurs van de Faculteit CITG hebben dagelijks een inlooppreekuur van 12.30 tot 13.30 uur. Hier kun je terecht met korte vragen, voor het inlooppreekuur hoeft je geen afspraak te maken. Stel je vraag per mail Heb je een vraag die ook via de mail kan worden beantwoord, stuur dan een mail naar studyadvisors-ceg@tudelft.nl Maak online een afspraak Denk je dat je vraag langer dan 10 minuten in beslag zal nemen, dan kun je online een afspraak maken van maximaal 30 minuten. Selecteer datum en tijd en maak een afspraak, je kunt de beschikbaarheid van de studieadviseurs zien voor de komende drie weken. https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/organisatie/studieadviseurs-citg/
Exit Qualifications	1) Analyse water related projects on their societal relevance, consequences, alternatives and interdisciplinary links. 2) Compute flows in selected water systems, evaluate and predict responses to natural and anthropogenic changes 3) Make and present a business plan for a water related business 4) Define a research question and set up your own research 5) Learn about measurements and design a measurement device 6) Collect relevant data and literature for a research topic 7) Perform research and report in the format of a scientific article
Administration by the Faculty of	CITG
Administration by the Education of	CiTG
Intended for	BSc students TU Delft and other national and international Universities.
Expected prior Knowledge	Although courses are introduction courses, basic calculus knowledge is expected. Knowledge of ordinary differential equations, linear algebra and basics of fluid mechanics is a pre. Depending on the pre-education, it can be required that background information through self-study is obtained. Some courses will contain assignments or exercises that are performed through programming in Python or Excel.
Prerequisites Minor	Minimum of 80 ECTS of the bachelor.
Minor Coherence / Goal	The water world is broad, has many aspects and has a big impact on the society, economy and the environment. This is in general hardly realized. A first theme in the minor is to point this out. The second theme is about how to set-up and perform a project or research of ones own liking. Basic knowledge to do so is obtained in the introductory courses.
Minor content 2	People expect that, when opening a tap, clean water flows out, and that floods do not occur. This are, however, big challenges that require contributions from different disciplines. Water is a pleasure, but it can also a threat. The minor shows these aspects and many more. There are fieldtrips in surrounding countries where you encounter features you never realized. Ever visited a hydropower dam? Do you know the consequences widening of rivers? Do we manage our water in the cities properly and what about the wastewater? In the first quarter of the minor a broad knowledge base is created, addressing water issues at the international, national, regional and urban level. In the same quarter a field excursion is planned and together with water experts you will study interdisciplinary examples of hydraulic structures and water resources management. You will learn about climate adaptation, how to start a project, research or water related business. This all in the course waterworks in practice. Due attention is given to observing and measuring in the field or the laboratory in the course Measurements for Water. You will design and construct a measuring device, to understand the behaviour of water and to learn about the design of measuring devices. Other courses include water and climate, rainfall in the city and water system analysis. The minor is concluded with a research project that you can define yourself, under guidance of experts in field of the research.
Education Methods	Lectures, (group)projects, practicals
Minor Assessment	Exams, grading of group work or individual work and presentations. See for more details the study guide descriptions of the individual courses.
Maximum number of participants	35
Minimum number of participants	30
Minor Remarks / Schedule	Period 1 CTB3360 Water System Analysis 4 ECTS CT3367 Introduction to Water and Climate 4 ECTS CT3411-16 Water works in practice (start)* 3 ECTS CT3412-20 Measurement for water 4 ECTS

Period 2

CTB3366 Rainfall in the city 4 ECTS

CT3411-16 Water works in practice (continued) 2 ECTS

CT3016 Research assignment 9 ECTS

*) There is a fee for the 4 day field excursion approximately Euro 80,- pp

CT3016	Project Minor Water Research	9
Responsible Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this assignment the minor is finalized with a project or research. Students select a topic in the field of Water Management (Citg) or Hydraulic Engineering (Citg). They can define their own research or select from available topics in the respective Departments. The project or research is presented in a report and a final presentation.	
Study Goals	With this assignment the student shows he/she is capable of analyzing a water related problem, defining a research question, and presenting in a systematic way a solution for the research question. The student must be capable of: -) structurizing a complex research question -) defining required field experiments or computer algorithms -) consulting relevant literature -) performing research -) reporting in a relevant format	
Education Method	The assignment is an (individual) research or project guided by relevant staff of the participating departments. It is finalized with a report and oral presentation. The assignment can be performed with maximum two students per research or project, only if each contribution can clearly be distinguished	
Assessment	A report in a relevant format and a final presentation. Judgement: - Being capable of defining a reseach question(10%). - Being capable of defining relevance and state of the art based on literature(10%) - Being capable of defining required steps to perform the research(20%) - Being capable of synthesizing all information in answering the reasearch question.(10%) - Being capable of reporting in the required format (30%) - Being capable of evaluating the research results(20%)	
Expected prior Knowledge	See general admission requirenments for minors	
Academic Skills	Academic aproach: Systematic approach, using existing theory. Capability of performing research or design: Discover own affinty with reasearch or design. Understand basics for making a research or design, being able to defend, and perform. Discover new points of view, being able to defend results. Planning a research or project. Incoorporate original new points of view. Communicate: Making use of literature and being able to report in a standardized format Critically analyze own results and from others. Written and oral communications with peers Societal context: Being capable of involving societal aspects in choice of research or design.	
Literature & Study Materials	Lectures, Presentations, Research examples	
Judgement	- Being capable of defining a reseach question(10%). - Being capable of defining relevance and state of the art based on literature(10%) - Being capable of defining required steps to perform the research(20%) - Being capable of synthesizing all information in answering the reasearch question.(10%) - Being capable of reporting in the required format (30%) - Being capable of evaluating the research results(20%) For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	na	
Collegerama	No	

CT3367	Introduction to Water and Climate	4
Responsible Instructor	Dr.ir. M.A. de Schipper	
Instructor	Dr. M. Vizcaino	
Instructor	Dr. D. Wüthrich	
Instructor	Dr.ir. R. Gelderloos	
Instructor	T.T. Kettler	
Contact Hours / Week x/x/x/x	5x2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The basic elements of and the relation between water and climate are highlighted and further discussed together with their mutual coherence.	
Study Goals	After this course you are able to: *explain the different processes of the global water cycle. *describe the flows of water and sand in different riverine, coastal and ocean systems. *identify mechanisms of climate change and you will be able to explain the interplay of climate change, sea level, clouds, rainfall and future weather. *explain why, when and which engineering interventions are needed in rivers, coast and urban environment. *explain why water for food and water for cities are the main challenges in water management and what the possibilities and limitations of reservoirs and groundwater are to improve water availability. *explain the challenges in better understanding and adapting to the impact of climate change on water for the coming 50 years.	
Education Method	Online lectures, texts and assignments. And, on-campus discussion and guest lectures during Education Period 1	
Computer Use	Yes, online content	
Assessment	Oral exam Judgement: During the oral exam the student is judged on his/her knowledge on the subject as well as the ability to discuss the matter in a broader context.	
Enrolment / Application	Register on brightspace	
Tags	Water Engineering Water management	
Expected prior Knowledge	High school level mathematics and physics.	
Academic Skills	Thinking (critical, analytical) Interpretation Cooperation Reasoning/arguing Awareness of and reflection on and responsibility towards the social (international) context and consequences of technology and scientific actions	
Literature & Study Materials	Study Material of the MOOC 'Introduction to Water and Climate' (online) + Material of the (on-campus) guest lectures	
Judgement	During the oral exam the student is judged on his/her knowledge on the subject as well as the ability to discuss the matter in a broader context. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Pocket calculator	
Collegerama	No	

CT3411-16	Waterworks in Practice	5
Responsible Instructor	Dr. E. Mostert	
Instructor	E. van Anel	
Instructor	A.W. Baar	
Contact Hours / Week x/x/x/x	0.2.0.0 week 1.1 t/m 1.7 0.8.0.0 week 2.1, 2.3, 2.7	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Globally, water challenges are on te increase, especially in vulnerable delta areas. How do we keep this areas safe and attractive to live in? How do we ensure sufficient water for producing food? The Dutch water sector has the ambition to contribute to the solution of these problems worldwide. In water works in practice, you will learn about these water challenges, the water works to address them, the water professionals and the water users, and you will think of different ways to valorise (increase the relevance and use of) water expertise in water management.	
Study Goals	After following the course, students will be able to: - Provide an overview of the different water managers and their roles - Provide an overview of the different water users and their needs and wishes - Provide an overview of the different water works and explain their function - Identify and analyse the main challenges in water management - Develop and validate a plan to valorise specific expertise for water management	
Education Method	The course consists of three parts/ In the first quarter you will get an introduction, via lectures, assignments and short field visits, of the field. Which parties are active? What are their roles? And what are the biggest challenges in the Netherlands and abroad? Als in the first quarter there is 4-day excursion to Limburg and its environs and to the deltaworks. You will visit water works and meet water managers. Moreover, you will take hydrological measurements and test measuring equipment you have made in the course Measuring water. The second quarter is about innovation. Innovation can be realized by applying new knowledge, by startup companies or existing companies, by the eater managers and water users themselves or by a combination of these. In this quarter a series of workshops will be given, where you are guided and helped to develop and validate a valorisation plan to implement a specific innovation in practice.	
Assessment	Provided each assignment is assessed with a 6 or higher, the final mark of the course is calculated as the weighted average of the marks of all assignment, the weight of all Q1 assignments together being 0.6 and of all Q2 assignments together being 0.4.	
Tags	Adventurous Fieldwork Group work Water management	
Expected prior Knowledge	Entry requirements minor	
Academic Skills	Analytical skills. Critical skills.	
Literature & Study Materials	Relevant materials will be posted on Brightspace. Moreover, students will have to find relevant materials themselves as part of the assignments.	
Judgement	To obtain a final mark for the course, every assignment should be sufficient (be marked 6 or higher after rounding off to the nearest half mark). The meaning of the marks for the different assignments is as follows: 5 or lower: does not meet the minimum requirements for the assignment 6: meets the minimum requirements, but only just 7: more than meets the minimum requirements 8: fully meets, but does not exceed, the requirements that can be expected given the available time and the background knowledge of the students 9: exceeds the requirements that can be expected given the available time and the background knowledge 10: outstanding Half marks indicate performance levels between two descriptions. Provided each assignment is assessed with a 6 or higher, the final mark of the course is calculated as the weighted average of the marks of all assignment, the weight of all Q1 assignments together being 0.6 and of all Q2 assignments together being 0.4. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CT3412-20	Measurements for Water	4
Responsible Instructor	Dr.ir. R.W. Hut	
Instructor	Dr.ir. A.M.J. Coenders	
Contact Hours / Week x/x/x/x	6 x 8 hours practical in total	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course deals with typical measuring techniques for the field of hydrology, water resources management and hydraulic engineering (water quantity). Topics are: get acquainted with different measuring techniques and instruments, accuracy and precision, designing a measuring plan (including sampling time) and safety. Students also learn how to execute measurements them self. During the workshop, students have to design, built, and test their own sensor for a 'water related' phenomena.	
Study Goals	1. Know relevant observation techniques for measuring water-related processes (quantitative and qualitative) in the rural and urban water cyclus. 2. Being able to select suitable observation techniques considering assumptions, shortcomings and application field 3. Determine the measuring frequency of an observation technique based on knowledge of the physical (water process) and order of magnitude of the process 4. Design and construct a mechanical and/or digital filter to observe high and low frequency water processes. 5. Design and construct a self developed observation technique for a water related process with sensors and electronics 6. Critically reflect on the self developed observation technique (applicability, accuracy, shortcoming, assumptions)	
Education Method	Lectures and workshop	
Assessment	design assignment (demonstration, instructable and report) all pass/fail.	
Expected prior Knowledge	none	
Academic Skills	design	
Literature & Study Materials	presentations, and hand-outs	
Judgement	workshop (demonstration, instructable, report)	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	
Collegerama	No	

CTB3360	Water System Analysis	4
Responsible Instructor	Dr.ir. G.H.W. Schoups	
Contact Hours / Week x/x/x/x	4/0/4/0 Hoorcollege: 2 Werkcollege: 2 Practica: 0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	Fluid mechanics Introduction water management Introduction hydrology	
Course Contents	Central topic of the course is analysis and design of natural and man-made water systems (canals, rivers, agricultural fields, polders, irrigation systems, soils, aquifers) based on hydraulic and hydrological knowledge. Common flow processes encountered in practice are discussed, including open-channel flow, hydraulic structures, soil water flow, and groundwater flow. In particular, the course focuses on: (1) unified treatment of the physics of water flow in surface and subsurface systems based on the principles of mass and momentum conservation (2) basic numerical methods for solving the relevant flow equations (3) application of theory to existing systems from around the world. As such, the course consolidates and builds on prior knowledge obtained in other courses, and prepares for more advanced topics in the MSc Watermanagement program.	
Study Goals	The overall goal is to analyze and quantify water flow through natural and man-made systems such as canals, rivers, agricultural fields, polders, irrigation systems, soils, and aquifers. This requires the following: quantitative understanding of flow processes commonly encountered in practice, including open-channel flow, hydraulic structures, soil water flow, and groundwater flow translate a given practical flow situation into a mathematical model of water flow in the system apply basic numerical techniques for solving the resulting mathematical models interpret computed results and draw conclusions about the functioning of the water system under consideration, and, if applicable, formulate control actions to improve the situation Study load: Lectures: 7 x 2 hours Assignments: 6 x 10 hours Self study: 7 x 2 hours Exam preparation: 8 hours Final exam: 3 hours	
Education Method	Lectures and Assignments	
Course Relations	Water management, Hydrology	
Literature and Study Materials	All study materials (lecture slides) will be provided on BrightSpace	
Assessment	Weekly computer assignments and final exam, with a split of 40/60 or 60/40, whichever yields the higher final grade. You need a grade of 5.75 or higher on the exam.	
Permitted Materials during Tests	Calculator and A4 paper with formulas prepared by the student	
Expected prior Knowledge	Fluid mechanics Introduction water management Introduction hydrology	
Academic Skills	Analytical thinking Report writing Problem solving	
Literature & Study Materials	All study materials (lecture slides) will be provided on BrightSpace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator and A4 paper with formulas prepared by the student	
Collegerama	No	

CTB3366	Rainfall in the City	4
Responsible Instructor	Dr. M.A. Schleiss	
Instructor	C.M.H. Unal	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr.ir. A.M. Droste	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- Urban pluvial flooding: costs, risks and trends. - Characteristics of flood producing rainfall systems - In-situ measurements of rain: rain gauges and disdrometers - Opportunistic rainfall measurements: citizen gauges and microwave links - Introduction to weather radar - Quality control, adjustment and merging of radar with ground observations - Rainfall extremes, return periods, IDF curves, design storms - Introduction to urban hydrology, rainfall runoff, infiltration, peak flow	
Study Goals	After this course, students should be able to: 1. Explain why cities are vulnerable to flooding and what is at stakes. 2. Examine the main techniques available for measuring rain and the principles on which they are based. 3. Discuss the challenges underlying rainfall measurements and forecasting in an urban setting. 4. Apply statistical methods to model rainfall extremes, calculate return periods, build design storms and detect trends in rainfall patterns using a simple programming environment like Matlab or Python. 5. Analyze the effects of rainfall and land cover data resolutions on hydrologic response. 6. Critically assess the usefulness and cost-effectiveness of current infrastructures and early warning systems for protecting cities from pluvial floods. 7. Summarize the results of published scientific studies in the form of a clear and concise written report.	
Education Method	Lectures and assignments	
Assessment	The assessment consists of 3 written assignments (70% of final grade) and one final written exam (30% of final grade). The grade for each part must be at least 5.	
Expected prior Knowledge	- Basics in programming (e.g., Matlab, Python or R) - Basics in statistics and probability theory - Calculus - Linear algebra	
Academic Skills	- Use of English language - Report writing - Critical reading - Self-management	
Literature & Study Materials	All necessary study materials will be provided on Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No materials will be allowed during the final exam except for: - a ruler - a simple calculator	
Collegerama	No	

CTB3366 Toets 1	Practical	2.8
Responsible Instructor	Dr. M.A. Schleiss	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Check CTB3366	
Study Goals	CTB3366	
Education Method	CTB3366	
Assessment	CTB3366	
Expected prior Knowledge	CTB3366	
Academic Skills	CTB3366	
Literature & Study Materials	CTB3366	
Judgement	CTB3366	
Permitted Materials during Exam	CTB3366	
Collegerama	No	

CTB3366 Toets 2		Exam	1.2
Responsible Instructor	Dr. M.A. Schleiss		
Contact Hours / Week x/x/x/x	0/6/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Course Contents	CTB3366		
Study Goals	CTB3366		
Education Method	CTB3366		
Assessment	CTB3366		
Expected prior Knowledge	CTB3366		
Academic Skills	CTB3366		
Literature & Study Materials	CTB3366		
Judgement	CTB3366		
Permitted Materials during Exam	CTB3366		
Collegerama	No		

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-170-24

Responsible Program Employee	D.M.S. Wevers-de Koning
Contact for students	Exchange-citg@tudelft.nl, room HG2.73
Program Title	Free Minor Study Abroad
Director of Education	T.J. Zitman
In association with the Faculty of	CITG
Prerequisites	a. students in their second year can apply for a minor abroad (in their third year) if they have a positive BSA (a postponed BSA doesn't qualify) upon submitting their exchange nomination package; b. students who apply for a minor study abroad in their 3rd year (or later) of registration, should have received at least a positive BSA and gained in total min. 75 EC in the BSc, upon submitting their exchange nomination package; c. to qualify for a minor abroad, students need to comply with the concerned administrative procedure, see the exchange manual of the year concerned. Further depending on the course package chosen.
Program Goals	A BSc-student is required to follow a consistent package of courses within a specific theme worth 30 EC. The course package may broaden or deepen student's knowledge and may not overlap the BSc major and your future MSc programme. The level of the course package should lead to 3rd year BSc level. Student studies 1 semester at a foreign partner university. Besides student experiences the way of living and studying in another cultural environment and expands his/her foreign language skills. All conditions for the free minor are to be found on the following website: https://www.tudelft.nl/en/student/ceg-student-portal/education/bachelor/self-composed-minor-information The package of courses is composed by the student him-/herself and needs to be officially approved by the free minor committee, before the study period abroad. This is part of the exchange nomination process. Please mind the following (BSc-students): it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Abroad only if student is studying nominally. A MSc-student is required to follow courses worth min. 20 EC after consultation with the Master coordinator. Grades obtained at the partner university will be converted to pass/fail. On the diploma supplement the original grades will be visible.
Exam requirements	BSc-student: the package of courses is composed by the student him-/herself and needs to be officially approved by the free minor commission CIE/AES, before the study period abroad. Student fills in the form 'Application Approval Free Minor' and asks a lecturer, specialized in the theme of the minor, to sign this form as well showing his/her support as far as the consistency of the course package is concerned. After the study period abroad, student is required to hand in the official transcript of grades to the Board of Examiners CeG as soon as possible. MSc-student: Check article 3C of the TER Annex of your study and starting year. This article describes how to go about electives in your study programme. Consult your MSc coordinator for approval of your proposed courses. The MSc coordinator has to approve your MSc curriculum by signing the AES/CIE-2 form. For TIL and CME students different rules may apply.
Intended for	CIE-students 5th semester BSc program, en MSc-students CIE. Please mind following (BSc-students): it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Study Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Study Abroad only if student is studying nominally.
Maximum number of participants	See the most updated overview with exchange places CeG available on Brightspace (Student and International Office CeG > Exchange).
Remarks Maximum number of participants	Depending on the number of exchange places available which may differ per partner university and per study year.
Minimum number of participants	0
Minor Remarks / Schedule	As from the beginning of the study year the Student and International Office CeG (SIO) will organize exchange information meetings. Please keep yourself informed of the data and other news by reading Brightspace and/or the announcements on the publication boards of SIO. All available exchange places CeG for both semester 1 and 2 and for both BSc and MSc students CeG will be set out through 2 nomination rounds. Please enroll Brightspace to keep yourself updated, and visit the exchange information meetings! The exchange manual which can be found on Brightspace is part of the exchange procedure so read this manual carefully! Please mind following: it is the responsibility of the student him-/herself to gain 30 EC for the minor abroad. If student does not succeed, student needs to take care of a suitable supplement in Holland to complete the Free Minor Study Abroad. This may lead to study delay. Therefore, it is strongly advised to participate in the Free Minor Study Abroad only if student is studying nominally. At the foreign partner universities different application deadlines and -conditions are applicable. Please keep in mind your own specific time planning. It is often not allowed to follow courses other than CeG at a foreign university. If you wish to follow courses in a completely different study area to broaden your knowledge, you need to contact the partner university yourself and ask for written permission for this. This may affect scholarship options. Students who spend an exchange period abroad, are required to send SIO digital experience reports during and after the exchange stay, and to act like a TU Delft ambassador upon request of SIO. The experience reports will be published on the faculty study abroad website and/or Brightspace. These reports are different from the ones you need to write for your (scholarship) application! After return, students also commit themselves to actively participate in the buddyprogram for incoming international students Finally it is the full responsibility of the student to start the exchange- and application procedure properly and in time and to meet the prerequisites and language requirements from the partner university in time as well. If we notice that during any moment of the exchange procedure (before and after nomination) you do not show a proper and flexible attitude towards the procedures, the Student and International Office and/or the host university keep the right of cancelling your nomination and to

not support your exchange study any longer.

Student needs to keep him-/herself updated of the latest exchange information, provided by exchange information meetings, the exchange manual and other communication channels like www.studyabroad.tudelft.nl, the faculty study abroad website, Brightspace.

No correspondence is possible about the outcome of f.e. the exchange nomination process. No withdrawal of the nomination (package) is possible.

Links:

www.studyabroad.tudelft.nl

www.studyabroad.citg.tudelft.nl

www.brightspace.tudelft.nl > Student and International Office CeG > Exchange

Year

Organization

Education

2024/2025

Civil Engineering and Geosciences

Minors Civil Engineering

CT-MI-174-24	
Minor Coordinator	J. Ninan
Contact for students	Minor mailbox: Minor-PM-CiTG@tudelft.nl
Director of Education	(CITG) Ir. T.J. Zitman
Program Coordinator	Dr. Johan Ninan

CT3101	Project Management Basics	5
Responsible Instructor	Dr. Ir. R. Kuttantherappel Soman	
Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is an important cornerstone of the minor Project Management: from Nano to Mega. In this course the general PM principles will be taught such as: iron triangle, trade-offs, scoping, scheduling, risk management, monitoring and control, project start-up, project close-out. Also the more strategic context of a project in an organization is discussed including portfolio management and the business case of a project. Throughout, the project life-cycle acts as a basis: A. Conception B. Definition C. Execution, contracting, implementation D. Handover Trade-offs between time, cost, quality, and scope are extensively discussed, after discussing the different axes in detail: -SCOPE axis - setting the project boundaries, WBS; -TIME axis - planning & scheduling, network scheduling, critical path method (CPM), Program Evaluation and Review Technique (PERT), resource constraints; -COST axis - estimating, project cost accounting, scheduling and forecasting for costs; -QUALITY axis - risk, TQM, six sigma, other techniques. Attention is paid to both front-end development and project execution. During the project execution the execution team utilizes all the schedules, procedures and templates that were prepared and anticipated during prior phases. Unanticipated events and situations will inevitably be encountered, and the Project Manager and Project Team will have to deal with it (change management). Overall: People are key in successful project management!	
Study Goals	The course is aimed at getting to know the basics of project management and being able to apply the obtained knowledge on real life cases. More specifically, after following this course the student is able to 1. Explain the difference between projects, programs and portfolios. 2. Explain how value is developed in projects throughout the project life cycle (PLC). 3. Select and reflect upon an appropriate organisation structure for a project. 4. Construct a project planning using MS Project /PertMaster. 5. Create a risk register based upon a qualitative risk analysis. 6. Explain the role of change management techniques in project management.	
Education Method	This course will be mainly based on lectures (/guest lectures) with smaller assessed weekly assignments for elaborating on the various topics. We will also use study materials from the MOOC Management of Engineering Projects: Preparing for success!	
Course Relations	This course is part of the minor Project Management: from Nano to Mega (CT-MI-174). Also students of the minor Offshore Wind Energy follow this course. Non-minor students might participate as well, but always should contact the lecturer upfront.	
Assessment	The assessment consists of - Digital exam (half knowledge questions, half questions related to a casus) - Assignments (obligatory), further specified during lecture 1, which all should be completed before the final grade of the exam can be obtained. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently).	
Tags	Business Project planning / management	
Expected prior Knowledge	No specific prior knowledge required.	
Academic Skills	This course will develop your writing and reflecting skills. And of course your project management skills!	
Literature & Study Materials	Management of Engineering Projects - People Are key. Hans L.M. Bakker and Jaap P. de Kleijn, editors. NAP - The Process Industry Competence Network. ISBN/EAN: 978-90-812162-0-3 This book can be ordered (student-price) via NAP or via the minor-staff (will be communicated in lecture 1).	
Judgement	The weekly assignments will be graded with fail - sufficient - good. All weekly assignments need to be at least sufficient: for a "fail", rework has to be done. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No materials are permitted during the Exam, except Dutch-English (or vice versa) dictionaries. Also a simple calculator can be used.	
Collegerama	Yes	

CT3102-15	Introduction to project finance & legal aspects of projects	5
Responsible Instructor	mr. J.M. Kooijman	
Instructor	Ir. M. van Os	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course will focus on the financial and legal aspects of projects and it consists of 2 parts; CT3102-15 Exam 1 (Legal Aspects of Projects) and CT3102-15 exam 2 (Financial Aspects of Projects). CT3102-15 Exam 1 (Legal Aspects): This course focuses on the legal aspects of projects both at the level of the projects, and at the level of the companies that undertake these projects. The course covers a wide variety of legal topics including new contracting and procurement methods, administrative law and environmental laws and regulations. CT3102-15 Exam 2 (Financial Aspects): This course will explain how investors and lenders determine whether to fund a project. A.o the following topics will be covered: project finance (=finance at the level of the project; not the stakeholders), financial analysis, optimising the financial structure, financial modelling and project risk management. Because any kind of project needs financing, and because they all need to follow the applicable laws and regulations, these courses are relevant for, and open to, students of all faculties at TU Delft. These courses are part of the minor Project Management: from Nano to Mega (MOT-MI 174). Non-minor students may participate as well but should always contact the lecturer upfront.	
Study Goals	Legal Aspects: After following this course, the student: - Knows the basics of contract law and different contract types, - Knows current project tendering procedures, - Knows the basics of administrative law, - And has some knowledge about the legal aspects of projects, including environmental rules and regulations. Financial Aspects: By the end of the course students should be able to: - Identify the financial analysis tools that project lenders and investors use to assess projects, and thereby predict their wishes and needs - Recognize how the optimal financial structure can contribute to the success of the project financing - Recognize how optimal contract terms can help a project to succeed, and illustrate how these support the project financing - Recognize why a risk management plan is vital to a successful project financing	
Education Method	This course will be mainly based on lectures with several assignments for elaborating on the various topics.	
Course Relations	This course is part of the minor Project Management: from Nano to Mega (MOT-MI-174). Non-minor students might participate as well but always should contact the lecturer upfront.	
Assessment	There will be 2 partial exams for this course: CT3102-15 Exam 1 (about the legal aspects) and Exam 2 (about the financial aspects of projects). Both partial exams count for 50%. The average score of Exam 1 and Exam 2 should be 6 or higher to pass the course. Due to existence the 2 different sub-codes, students can carry forward a partial grade for a sub-course course for averaging with a later grade in the parallel course. Please note that for carry forward a grade 5 or higher is required. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows : timed take-home exam or timed take-home assignment.	
Tags	Analysis Business Economics Finance Modelling Project Project planning / management Projects	
Expected prior Knowledge	There are no formal prerequisites for this course.	
Academic Skills	Through taking this course, students will aid the development of the following academic skills: - Moral awareness/sensitivity - Judgemental skills - Professionalism - Integrity - Reasoning/arguing - Logic	
Literature & Study Materials	For the sub-course Financial Aspects, the course book will be the book Principles of Project Finance by E.R. Yescombe, 2nd edition, ISBN 978-0-123-91058-5. More information on other study materials will follow via BS	

Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations There will be 2 partial exams for this course: CT3102-15 Exam 1 (about the legal aspects) and part B (about the financial aspects of projects). Both partial exams count for 50%. The average score of Exam 1 and Exam 2 should be 6 or higher to pass the course. Due to existence the 2 different sub-codes, students can carry forward a partial grade for a sub-course course for averaging with a later grade in the parallel course. Please note that for carry forward a grade of 5 or higher is required. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.
Permitted Materials during Exam	No books and lecture slides are allowed, switch off cell phones and put them out of sight. Unprogrammed calculators are allowed, as well as English-Dutch dictionaries.
Collegerama	No

CT3102-15 Toets 1	Legal Aspects	2.5
Responsible Instructor	mr. J.M. Kooijman	
Contact Hours / Week x/x/x/x	See CT3102-15	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	For the course information see 'CT3102-15'	
Study Goals	See CT3102-15	
Education Method	See CT3102-15	
Assessment	See CT3102-15	
Expected prior Knowledge	See CT3102-15	
Academic Skills	See CT3102-15	
Literature & Study Materials	See CT3102-15	
Judgement	See CT3102-15	
Permitted Materials during Exam	See CT3102-15	
Collegerama	No	

CT3102-15 Toets 2	Project finance	2.5
Responsible Instructor	Ir. M. van Os	
Responsible Instructor	mr. J.M. Kooijman	
Contact Hours / Week x/x/x/x	See CT3102-15	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See CT3102-15	
Study Goals	See CT3102-15	
Education Method	See CT3102-15	
Assessment	See CT3102-15	
Expected prior Knowledge	See CT3102-15	
Academic Skills	See CT3102-15	
Literature & Study Materials	See CT3102-15	
Judgement	See CT3102-15	
Permitted Materials during Exam	See CT3102-15	
Collegerama	No	

CT3103	Integration: technical project	10
Responsible Instructor	Dr. E.J. Houwing	
Instructor	J. Ninan	
Contact Hours / Week x/x/x/x	6/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In course CT3103 students work together in small teams (6 students) consisting of students with different BSc backgrounds, thus enabling real integration of knowledge. They will apply their knowledge of processes and products in a realistic project setting, whereas the interaction between project and environment is a central theme. The students learn how to manage an integral project, how to lead a group of students (process- and project management), work together, develop plans, collect data, how to communicate on the project and how to present the intermediate and final results to external parties in a professional way. The goal is to practice skills in (simulated) practice settings and apply the project management methods and techniques from other courses in the minor. As a result, our project education is about the acquisition of knowledge and competences that are not acquired elsewhere in the curriculum, and has two learning objectives: LO1: After this course students are able to apply new knowledge on management, business finance, collaboration and legal aspects in order to produce a well thought plan for a good design. LO2: After this course students are able to evaluate their design on applicability, maintenance and acceptance by stakeholders and the client.	
Study Goals	Competence in: - Identification of user needs for a large building project; - Development of a project proposal; - Development of a business plan for a construction company; - Development of a design proposal in which multiple disciplines are integrated and applied; - Preparation of an integral Design (tender) based on different requirements/stakes brought up by the different stakeholders. The final Design is based on realistic construction and financing of the building object; - General project management skills, including teamwork, organization, management, fit for purpose, presentation and report writing.	
Education Method	- Project education; the central assignment is the development of a plan for a multidisciplinary building project. - Supporting lectures and workshops on dedicated subjects such as teamwork, project finance, legal aspects, and BIM, as well as guest lectures by project managers from practice.	
Assessment	The final mark for this course is a weighted average of marks for the following parts: (1) Project Proposal & Business Plan (2) Design (including Project management Plan, preliminary Design) (3) Construction (including concept proposal for final Design and Execution) The resulting group mark can be adjusted for individual students based on peer review results.	
Expected prior Knowledge	Third year University level in engineering, design, planning, project management or similar	
Academic Skills	Collaboration in teams, reflection on collaboration, oral presentations, writing reports, awareness of societal impact of large construction projects	
Literature & Study Materials	Will be made available through Brightspace	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No written exam	
Collegerama	No	

CT3104	Fit-for-purpose projectmanagement	5
Responsible Instructor	Ir. M. van Os	
Responsible Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	More and more a contingency approach in project management is advocated in literature: rather than just applying basic project management theory for all projects (regardless specific project characteristics), project management should be tailored to suit a particular project. This course will focus on this fit-for-purpose project management. Principles of project management taught in this minor will be treated using cases from project management practice in various industries and public environments. The lectures are taught by staff that have a combined experience from industry and academia as well as by guest lecturers. In this course several open-ended cases will be treated, for which student teams will have to find specific solutions. Apart from the size and the shape of the project, each project manager has a number of specific preferences for managing his project. The presenters will show what fit-for-purpose means to them in their specific project based on their own 10 dos and don'ts for project management. Preferably the presenters will deliver best practices per phase, per company or per project type and it is up to the students to make their own balanced selection. The students will gather over the various lectures a set of best practices and by comparing and challenging the presenters will come up with a fit-for-purpose approach for their own project (same project as in the integration course, course 3, different assignment). In this way they deliver a set of minimum standards for future reference and are able to challenge future presenters on their experience. Site visits are part of this course.	
Study Goals	The student - Knows various ways to adapt projects in a fit-for-purpose way - Can apply fit-for-purpose project management - Is capable of critically reflecting on the need for project management in a wide variety of domains - Recognizes the flexibility of project management.	
Education Method	The lectures are taught by staff that have a combined experience from industry and academia as well as by guest lecturers. In this course several open-ended cases will be treated, for which student teams will have to find specific solutions	
Assessment	6 essays throughout the course and application of the learnings to their integration project as well as a final article.	
Expected prior Knowledge	Judgement: Based on Rubric which will be shared with the students.	
Academic Skills	CT3101	
Literature & Study Materials	Writing. Literature survey. Analysis and application of the learned.	
Judgement	Presentations and supporting articles.	
Permitted Materials during Exam	Based on Rubric which will be shared with the students.	
Collegerama	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
	NA	
	No	

TBM610A	Process Management and Decision-making in Project Environments	5
Module Manager	Prof.dr. W.W. Veeneman	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. H.G. van der Voort	
Instructor	Dr. M. Leijten	
Instructor	Dr.ir. E. Minkman	
Responsible for assignments	Prof.dr. W.W. Veeneman	
Co-responsible for assignments	Dr. M. Leijten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Projects are generally managed with a strong focus on predicting the schedule, budget, scope resources and tasks. That phase of predicting is then followed by a strong focus on controlling the works towards the predicted outcome (predict and control). However, many projects have to deal with significant uncertainties, leading to surprises in the execution of the project. Process management offers a different view on projects, focussed on dealing with surprises by preparing for them and committing parties to the project (prepare and commit). It can help steer the project from separated task orientation to an integrated goal orientation. The course introduces the students to the way in which process management can be used in projects and how that translates to contracts, structures and day-to-day management. This course is part of the minor project management. (Please contact the module manager in case you would like to follow this course outside the minor program).	
Study Goals	Understand how complexity and uncertainties reveal themselves in projects and how to be prepared for them. Understand the role and forms of process management in managing complex projects, as an addition to standard project management tools.	
Education Method	After a number of introductory lectures the students write an essay in small groups on how to apply process management in a situation of their choice, based on recent news events on large engineering projects. Their product is a process design and approach plan.	
Assessment	The students are asked to write an essay on the matter using literature and a few interviews. After that they are themselves interviewed on their report. Their grade is for 80 percent the report and for 20 percent their interview grade	
Elective	Yes	
Category	BSc level	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-186-24	
Minor Coordinator	Drs.ing. J.J. de Boer
Program Coordinator	Drs.ing. J.J. (Hans) de Boer.
In association with the University of	Guest lectures by scientists and practitioners from engineering firms, municipalities, province and ministry.
Program Start (Study Year)	2024/2025.
Administration by the Education of	Faculty of Civil engineering and Geosciences (CeG).
Minor Title	Integrated Infrastructure Design (IID).
EC Minor (Volume)	30.
Minor Semester	1.
Academic Chair or Topics involved	Integral Design and Management, Department of Materials, Mechanics, Management & Design -3Md- (CeG).
Cooperating Education	Faculty of Architecture and The Built Environment (Arch), Faculty of Civil Engineering and Geosciences (CeG), Faculty of Technology, Policy and Management (TPM).
Cooperating Academic Chair	Urbanism: Urban Design, Landscape Architecture; Architecture: Theory & Territories, Complex Projects; Architectural Engineering + Technology: Structural Design and Mechanics; Materials, Mechanics, Management & Design (3Md): Integral Design & Management, Applied Mechanics; Transport and Planning; Engineering Systems & Services: Transport and Logistics.
Contact for Students (Minor)	Drs.ing. J.J. (Hans) de Boer, j.j.deboer@tudelft.nl.
Intended for	Dedicated and motivated BSc students or students from universities of applied sciences could further their knowledge of transport and water infrastructures, its integrated design and to develop skills on design, (3D) drawing & modeling, communication, presentation and multidisciplinary collaboration. This minor confronts you with (inter)national infrastructural challenges for the next decades. For architecture and civil engineering students in particular the minor offers actual assignments from practice which familiarize you with issues and demands relevant for your future professional orientation.
Minor Exit Qualifications	Students who have accomplished this minor successfully can further their ambitions in becoming a designer/engineer: who understands and who can reflect on the notion of integrated design for transport and water infrastructures and its added value for society; who understands and who is capable to apply notions and perspectives from different disciplines; who is capable of (re)defining and analyzing design issues and translate these in design objectives and criteria within a technological, environmental, societal and economical context; who is creative to develop design alternatives and who can give design decisions a solid ground on basis of a multi-criteria analysis and convincing arguments; who can present an integrated (3D) design with an attractive architectural expression and with elaborated technical details; who is capable to work methodically within a multidisciplinary team in a studio setting. And at the end who is capable to design an innovative and attractive infrastructural object such as a bridge, tunnel, viaduct, dike, station or other civil infrastructure which is well integrated within the transport and/or water system and within the urban fabric and/or natural environment.
Minor Coherence / Goal	This minor offers students a comprehensive set of interrelated courses on integrated design of transport and water infrastructures such as stations, bridges, tunnels, viaduct, dikes, roads and railways. The minors title - integrated design - means that courses focus not only on technical challenges such as function, construction and materials, but also on its architectural form and expression, its integration in the transport system, and its spatial integration within the urban fabric or natural environment.
Minor Content 1	The minor starts with an introduction to various integrally-designed transport and water infrastructures, their historical developments and environmental context, and to a range of design perspectives and approaches to integrated infrastructural design from theory and practice. Students have to write an essay on basis of individual research for an infrastructural object or aspect of their own choice. During the introduction there will be also an excursion on transport infrastructures and an excursion on water infrastructures. (Course BK7932).
Minor content 2	Subsequent courses delve into transport and water infrastructures from various perspectives and aspects such as infrastructure planning and governance (Course CT3200) and the idiosyncrasy of (infra-)structures as an object/artefact (Course BK7933). The relationship of infrastructures and its challenges with contemporary urban and landscape design, spatial and technical conditions will be subject of the course dedicated to environment and infrastructures (Courses BK7935).
Minor content 3	All these courses have exercises, individually and by student teams, to get a better understanding of theoretic notions and to experience its application to different cases.
Design Project	To stimulate and to develop students' actual design skills in-depth a dedicated course with design exercises on different scales and with increasing complexity runs parallel to the more theoretic courses on basis of assignments from practice (Course BK7934).
Education Methods	All courses are preparing for the final project where students will work in a multidisciplinary team to resolve a challenging and actual infrastructure design issue from practice. This on basis of an assignment from a client from a real-world context which aims for an innovative and attractive integrated (re)design of a bridge, tunnel, viaduct, dike, station or other civil structures as part of a route within and between neighborhoods or crossing a waterway connecting larger urban areas which contributes to societal values such as accessibility, livability, social inclusiveness and economic competitiveness (Course CT3201).
Minor Assessment	Lectures, workshops, seminars, assignments, exercises, projects, excursions.
Remarks Maximum number of participants	Each course will be examined separately on basis of a written report, essay, design portfolio, or booklet combined with presentations depending on the course. The separate results must meet the minimum requirement of 6,0.
Minimum number of participants	30.

BK7932	Introduction to Integrated Infrastructure Design	3
Course Coordinator	Drs.ing. J.J. de Boer	
Course Coordinator	J.R.T. van der Velde	
Instructor	Drs.ing. J.J. de Boer	
Instructor	M. Triggianese	
Instructor	J. Ninan	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course gives an introduction to the main theme of the minor, presenting the historical, theoretical and methodological frame of integrated infrastructure design and its current manifestation and future design challenges from various perspectives. Transport and water infrastructures like roads, bridges, railways, stations, waterways and dikes form the focus of the course, both as physical structures as well its system and spatial integration. The course is presented via lectures, exercises, fieldtrips and selected literature. Content-wise, the course includes: - Key historical examples of integrated Infrastructures; - historical overview of transport and water infrastructures; - an overview of contemporary infrastructure and its integrative problems and potentials; - definitions, concepts and methods for integrated infrastructure design; - contemporary paradigms and positions in integrated infrastructure design; - hypothetical lenses for integrated infrastructure design;	
Study Goals	At the end of the course a student: 1. has a basic knowledge of the key historical examples of integrally designed Infrastructures and can reflect on these; 2. has a rudimentary understanding of the historical development of transport and water infrastructures and can reflect on them; 3. has a working understanding of the various types of infrastructures and its integrative problems and potentials; 4. can describe and distinguish between general definitions and concepts on infrastructure design and integrated infrastructure design; 5. can identify and describe contemporary paradigms and positions in integrated infrastructure design; 6. can reflect on various hypothetical lenses in relation to integrated infrastructure design challenges; 7. is capable to review critically an infrastructure and its integrality by clear reasoning and sound arguments on basis of a clear formulated research question, a well-chosen and applied appropriate theory and on basis of relevant information and data.	
Education Method	1. Instruction and feedback sessions on design approaches and methodologies, essay writing. 2. Lectures on knowledge aspects; 3. Exercises in groups on skill aspects; 4. Excursions to projects and study areas;	
Literature and Study Materials	Acker, van, M. (2014). From flux to frame: Designing infrastructure and shaping urbanization in Belgium. Leuven, Leuven University Press. Boomen, van den, T.et.al (2014). Onder weg!, Vijftien ontwerpen voor Transit Oriented Development (TOD) aan de Zaancorridor. Amsterdam, BNA Onderzoek. open access - Boomen, van den, T. et. al (2016). Highway x City, The future of the urban ring road. Amsterdam, BNA Onderzoek. open access - Hertogh, M., E. Westerveld (2010) Playing with Complexity, Management and Organisation of Large Infrastructure Projects (Doctoral Dissertation). open access - Kothuis, B, Kok, M. (eds) (2017). Integral Design of Multifunctional Flood Defenses, Multidisciplinary approaches & examples. Delft, Delft University Publishers. open access - Nijhuis, S., Jauslin, Van Der Hoeven, F.D. (eds) (2015). Flowscapes. Research in Urbanism Series, Volume 3. Delft: TU Delft & Delft Infrastructure and Mobility Initiative (DIMI) open access - Shannon, K., & Smets, M. (2010, herdruk 2016). The Landscape of Contemporary infrastructure. Rotterdam: NAI010. Sijmons, DF (2002). The Skill and the art of the Engineer, in Sijmons, D. and Ball, G. Landscape. Amsterdam: Architectura + Natura. Triggianese, M. (eds) (2018). Stations as Nodes: exploring the role of stations in future metropolitan areas from a French and Dutch perspective. Delft, Architecture and the Built Environment. open access - Velde, van der, J.R.T. (2018). Transformation in Composition, Ecdysis of Landscape Architecture through the Brownfield Park Project 1975-2015. Delft, Architecture and the Built Environment. open access - Verheijen, M. (2015). Infratecture: Infrastructuur als Ontwerppogave. Rotterdam: NAI010. Voorendt, M.Z. (2017). Design principles of multifunctional flood defences. Delft, TU Delft,- open access -. Wilbers, J. et. al (2013). Integraal Ontwerpen van Infrastructuren: aanzet tot een Agenda voor Onderwijs en Onderzoek. Delft, TU Delft, open access -.	
Assessment	The final product of the course is an individual written essay as critical review of a chosen infrastructure from a theoretic perspective and critical attitude. The following criteria will be considered: 1. Subject definition & research/review (sub)question(s) (focus and scope of position or review), max. 15 points. 2. Theory, concept, paradigm, lenses (choice, definition and argumentation), max. 15 points. 3. Logical structure & critical attitude (set-up and logical steps in the argument and critical attitude to the position or review of the object or subject from the theoretical and methodological notions), max. 40 points. 4. Final conclusions (on the position or review of the object or subject and in the context of integrated design), max. 15 points. 5. Presentation & references (structure and layout, use of language and writing skills, use and choice of sources), max. 15 points.	

Period of Education	At the level of learning objectives the essay should reflect students:
	1. Ability to reflect on integrated infrastructure design from theoretical and methodological notions presented in the course and to take a critical position on these;
	2. Authority in recognizing, reproducing, and challenging various types of infrastructures and its integrative problems and potentials.
	Week 1.1: Introduction Week 1.7: Progress-session Week 2.3: Final-session

BK7933	Idiosyncratic Infrastructures	3
Course Coordinator	Ir. F. Geerts	
Responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Course Contents	Complementing integrated design striving for a synthesis benefitting from a dialogue between disciplines, the course practises the inverse integrated analytical approach, using the particular disciplinary backgrounds of the students. The design of infrastructures and its component parts is dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. Idiosyncratic solutions have however often remained off-the-radar, and do not prominently contribute to the body of knowledge of infrastructure design. The course intends to remedy this knowledge gap, by collecting, retro-engineering, redrawing and categorizing pieces of infrastructures. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, eventually results in a catalogue of idiosyncratic infrastructures. The course which exists out of a series of seminars will be finalized by a poster-presentation and a portfolio of drawings documenting one infrastructural artefact.	
Study Goals	At the end of the course a student: 1. has a basic knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures and mentoring, within a studio setting. 2. Seminar tutorial format with student participation through class discussion and student presentations. 3. The drawing-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	poster, presentation, portfolio, class-participation, weekly progress On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (q2)	

BK7934	Design of (infra-)structures	6
Course Coordinator	Drs.ing. J.J. de Boer	
Course Coordinator	Dr.ir. J.E.P. Smits	
Contact Hours / Week	8 hours per week	
x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	An introduction to transport infrastructures. The course focuses on the design of a civil structure such as a bridge, a tunnel or a series of infrastructures. This as part of a route design through a landscape or as part of a traverse through a city representing a higher scale and system level. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design. Cross over and introductory workshops dedicated to environmental integration, mapping out roads and waterways, geometric road design and construction, hydraulic structures design and construction are part of this course. The course will be finished by oral and written presentations with drawings, sketches, schemes and models by teams of students based upon a case from practice presented by a commissioner for the design of transport infrastructures. This case will be aligned with the assignment for the final design project.	
Study Goals	The student: 1. Is capable to set up and execute a design process of an infrastructural assignment on different scale and system levels; 2. is able to derive and apply design criteria for the bridge or civil structure from the spatial context; 3. is able to derive and apply design criteria for the bridge or civil structure from technical specifications; 4. acquires knowledge of the different disciplines such as landscape design, urban design, architectural design and structural design and contributes from the own discipline to the design of a bridge or civil structure.	
Education Method	Lectures, workshop, studio work.	
Assessment	The presentations will be assessed on: 1. understanding of the integral design process (15%); 2. ability to make a good architectural and structural design that responds to its urban/landscape context (50%); 3. ability to translate technical specifications into a functional design (20%); 4. group work and integral cooperation (15%).	
Period of Education	Quarter	

BK7935	Environment & Infrastructures: Urban Systems + Landscape	6
Course Coordinator	Dr.ir. I.U. Hurkxkens	
Module Manager	Dr.ing. M.Z. Voorendt	
Instructor	Ir. G.J.M. van der Meulen	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1 2	
Start Education	1 2	
Exam Period	1 2	
Course Language	English	
Summary	This course develops core knowledge and skills around the topic of infrastructure in relation to various (spatial) environments. It deals with both the urban environment and non-urban realms such as cultural and natural landscapes. Given the overlaps and complementarities of these realms, the courses structured into three parts: Urban Systems, Landscape, & Integrated Project.	
Course Contents	A. Urban systems The first part of this course aims to train students in understanding, analyzing and assessing urban infrastructures as a part of complex urban systems. We will focus on three main systems: traffic, water (including flood defense systems), and green space/public space. The course will pay close attention to infrastructure zones in the cities which underwent transformations recently, and which have to adapt to a changing context in the future. We will consider three scales and their interrelations: the scale of the city as a whole, a zone of integrated infrastructures, and a specific section of a zone in detail. Starting at the city scale, we will zoom in to scale of the zone, followed by the scale of the section. Based on a proposal on the scale of the section, we will zoom out again to zone scale, and to the city scale, extrapolating the effects of the proposed intervention. While doing so the spatial and structural integration of the infrastructures will be taken into account. B. Landscape The second part of the course elaborates on the engineering and design considerations of integrated infrastructure design in a landscape context. It explores the technical principles of primary flood defense systems, the principles of route design in a landscape context, the formative role in infrastructure in the development of landscape, the perception of landscape environments from various speeds and modalities, and integrated infrastructure design at the different (landscape) scales. The focus is on Waterscapes (river and coastal flood defense, dikes, waterfronts) and Road-scapes (paths & cycleways, roads & highways). C. Integrated Project The course concludes with an exercise synthesizing the knowledge and skills developed in both parts of the course in a complex infrastructural project in a peri-urban context.	
Study Goals	At the conclusion of the course a student: - Has the ability to recognize and represent different urban systems; - Has the ability to understand the way different urban systems are related to each other and influence each other; - Has the ability to analyze and understand urban infrastructures as complex combinations of urban systems; - can indicate improvements to urban infrastructures as combinations of urban systems; - understands the historical and conceptual relationship between landscape and infrastructure; - has a working understanding of the engineering and design considerations of 'water-scapes in a landscape context'; - has a working understanding of the engineering and design considerations of 'road-scapes in a landscape context'; - can develop a schematic design for an integrated flood defense element incorporating the above skills;	
Education Method	Lectures, individual exercises, presentations, readings, discussions, group project work.	
Literature and Study Materials	Palmboom, F. (1989). Learning from Rotterdam. New York: Alexandrine Press Book. Dee, C. (2001). Form and fabric in landscape architecture: a visual introduction. Londen, New York: Spon Press. Shannon, K. & Smets, M. (2010). The Landscape of Contemporary infrastructure. Rotterdam: NAI Publishers, Rotterdam. Prominski, M., Stokman, A., Stimberg, D., Voermanek, H. & Zeller, S. (2012). River.Space.Design: Planning Strategies, Methods and Projects for Urban Rivers; Basel: Birkhauser Roozenburg, N.F.M.; Cross, N.G. (1991) Models of the design process: integrating the disciplines. Voorendt, M.Z. (2015). The 'Delft design method' for hydraulic engineering, Chapters 2, 4 and 5. Delft University of Technology / Bees Books, Amsterdam.	
Assessment	Individual assignments for the Urban Systems and Landscapes parts of the course. Teamwork for the Integrated Project: report & presentation, application design method, analysis, technical aspects, landscape aspects, reflection on urban systems aspects, reflection on integrality.	
Period of Education	Week 1.2 - 2.2	

CT3200	Infrastructure Planning & Governance	3
Responsible Instructor	Drs.ing. J.J. de Boer	
Responsible Instructor	Dr. J.M. Vleugel	
Instructor	Drs.ing. J.J. de Boer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The focus of this course will be on planning and governance of transport infrastructures. There is growing recognition that transport infrastructure plays a crucial role in the spatial and economic development of a region or country. The scope of infrastructure projects has increased, requiring to integrate a large number of aspects into the planning and design of transport infrastructure on all spatial scale levels, covering technical, financial, economic, and social aspects. In recent years there has been considerable innovation in the methods/approaches to plan and implement infrastructures. The course gives a kaleidoscopic overview of key topics in planning and governance of transport infrastructures and their interaction in theory and practice. Course topics will be: - Interaction between transport infrastructures and spatial-economic development; - Spatial planning and transport; - Design of transport infrastructures; - Infrastructure planning: technical approach, planning procedures, funding and procurement. - Impacts of transport infrastructure: accessibility, urbanization, social, economic, environmental, safety; - Assessment of infrastructure plans using tools like multi-criteria analysis, cost-benefit analysis, life-cycle costing.	
Study Goals	The main goal of the course is to provide student with basic knowledge, insights and analytical tools to plan transport infrastructure projects and networks. After passing the course the participant will be able to design his/her integrated spatial plan, or make a critical review of existing ones. Based on the many example projects the course will further provide the participant with a sense (combination of technical and governmental insight) for optimization of transport infrastructures.	
Education Method	Lectures by lecturers and professionals from planning practice. Group projects with autonomous team work. Group presentations about progress and final results. Defense of the results. Students should attend all lectures and presentations. Non-attendance has to be discussed with the instructor for that day before the specific lecture. The same goes for the weekly progress meetings.	
Prerequisites	The regular enrolment criteria for the minor apply.	
Assessment	The course will be finalized with a report by teams of students based upon a chosen case concerning the analysis and learnt lessons of the planning and decision making of existing infrastructure. The project, its results and the design process will be described in a concise report. Each report starts with a dedicated and focused problem analysis. The problem analysis finishes with a vision on the problem and design criteria. The latter will be translated into one or more plans. These plans will be evaluated. The final section of the report contains the conclusions and recommendations for follow-up research. The information for the report will be derived from lecture materials, literature references and desk research. The report reflects an awareness and use of all lecture materials as described above. The report contains not only text, but also visualizations. It should also be concisely written.	
Expected prior Knowledge	Relevant knowledge from earlier bachelor courses.	
Academic Skills	Able to apply methods for problem analysis, needs analysis, design and evaluation.	
Literature & Study Materials	John Baggen & Hans van Ham: Het Transportsysteem. Delft Academic Press, 2019. ISBN 9789065624406. Available for reduced price via university. There is no English version of this book.	
Judgement	Reports and presentations will be assessed using the set of objective criteria. The latter can be found on Brightspace. The final result is based on the quality of the report, the presentations, the discussion and final defense.	
Permitted Materials during Exam	No written exam.	
Collegerama	No	

CT3201	Interdisciplinary & Collaborative Design Project	9
Responsible Instructor	Drs.ing. J.J. de Boer	
Responsible Instructor	Prof.dr.ir. M.J.C.M. Hertogh	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course is the final project of the minor Integrated Infrastructure Design (IID) where the acquired theoretic knowledge and design skills of former courses will come together in an assignment for a challenging design issue. This design issue is related to several infrastructural interventions within the transport and/or water system and its environmental context and should result in an integrated infrastructure design. The assignment will be provided by an organization from the domain of infrastructures, mobility and environment where representatives will act in the role of commissioner. The focus of the design issue will on an infrastructural object or artefact like a bridge, tunnel, viaduct, dike, station or other civil structure which is characterized by its function, form, construction, materials and its architectural expression. This infrastructure should be well integrated within the transport and/or water system like a route, traverse or waterway and within the urban fabric and/or natural environment. The students will work autonomously in teams composed by different disciplinary backgrounds on basis of an project plan which must be set-up by the teams. Each team will work on a different design issue. The teams will (re)define the design issue and objectives, explore and analyze the environmental and societal stakeholder context for its possibilities, constraints and conditions. Lenses such as functionality, sustainability, spatial quality, perception and/or added value should be deployed as a guiding design principle. The findings and conclusions of the analysis must be translated, next to the functional and technical requirements of the (infra-)structures itself, into a coherent vision and related design objectives, specifications and criteria considering the involved scales and overall integration. The design alternatives and its final design should be evaluated against objectives, specifications and (multi)criteria and should be an expression of an integrated design. The final design and report will be assessed with respect to the way student teams were able to deal with the: (re)definition of the design issue and objectives; analysis including findings and conclusions; vision, design decisions and arguments; and ultimately with the synthesis/creation of an integrated infrastructure design which facilitates societal and economical needs and activities. This as a result of a collaborative and multidisciplinary design process which will also part of the assessment.	
Study Goals	At the end of the course a student: 1. has proven to be capable in applying the acquired knowledge and skills of the other courses of the minor into the final project; 2. is capable to (re)define and analyze a design issue for an infrastructural object considering different contexts and scales from the point of view of integration; 3. is capable to create an innovative and integrated infrastructural design for a bridge, tunnel, viaduct, dike, station or other civil structure regarding its functional, structural and architectural design and which is well integrated within the transport and/or water system and within the urban fabric and/or natural environment; 4. is capable to work methodically in a design project in collaboration with students from different disciplinary backgrounds.	
Education Method	1. Instruction and feedback sessions regarding assignment and design issue, project set-up and execution. 2. Sessions with lecturers from other courses of the minor for interpreting the design issue and objectives from different disciplinary perspectives. 3. Sessions with the commissioner and several experts from practice regarding the assignment and related developments, policies, information and data. 4. Autonomous design activities by the student teams within a studio setting.	
Assessment	Oral presentation and report by student teams with: (re)definition of the design issue and objectives; findings and conclusion of the analysis; vision and derived design objectives and criteria; multi-criteria analysis, design decisions and arguments; drawings, sketches, schemes and models with relevant details and measures. The lecturers each contribute with a sub-mark. The course coordinator determines the final mark based on the sub-marks and the five judgement criteria. Judgement: 1. Innovativeness and integration of the infrastructural design(s) at the functional, technical and architectural object level (30%); 2. Integration of the infrastructural design(s) at the transport and/or water system level (15%); 3. Integration of the infrastructural design(s) within its environmental context of the urban fabric and/or natural environment (15%); 4. The methodological approach applied to the design considering (re)definition, analysis, vision, objectives and criteria, conceptualization and alternatives, (multi)criteria evaluation, design decisions and arguments (20%); 5. Set up and execution of a project plan for the design project and the multidisciplinary collaboration within the project (20%).	
Expected prior Knowledge	Theory and skills from the other courses of the minor: BK7932, BK7933, BK7934, BK7935, CT3200.	
Academic Skills	Not applicable	
Literature & Study Materials	Dictaat CTB1220 - Integraal Ontwerp en Beheer, Faculteit Civiele Techniek en Geowetenschappen, Prof. dr. ir. M.J.C.M. Hertogh, Dr.ir. M.G.C. Bosch-Rekvelde, November 2016. Shannon, K., & Smets, M. (2010). The Landscape of Contemporary infrastructure. Rotterdam: NAI010 Publishers. Verheijen, M. (2015). Infractecture: Infrastructuur als Ontwerppogave. Rotterdam: NAI010 Publishers.	
Judgement	1. Innovativeness and integration of the infrastructural design(s) at the functional, technical and architectural object level (30%); 2. Integration of the infrastructural design(s) at the transport and/or water system level (15%); 3. Integration of the infrastructural design(s) within its environmental context of the urban fabric and/or natural environment (15%); 4. The methodological approach applied to the design considering (re)definition, analysis, objectives and criteria, conceptualization and alternatives, (multi)criteria evaluation, design decisions and arguments (20%); 5. Set up and execution of a project plan for the design project and the multidisciplinary collaboration within the project (20%). For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during	Not applicable	

Exam	
Collegerama	No

Year

Organization

Education

2024/2025

Civil Engineering and Geosciences

Minors Civil Engineering

CT-MI-188-24	
Introduction 1	Please be aware of the fact that this programme is only a port of the total bridging programme. After finishing your BSc, you need to do an additional programme of 32 EC.

CT3182	Module 5 Levenscyclus analyse	3
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Tags	Sustainability	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	No	

CTB1210	Dynamica & Modelvorming	5
Responsible Instructor	Dr.ir. A. Gavriilidou	
Instructor	Dr.ir. J. Bosboom	
Instructor	Dr. A.A. Núñez Vicencio	
Instructor	Dr. A. Cabboi	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	No	

CTB1310	Construction Mechanics 2	5
Responsible Instructor	Prof.dr.ir. J.G. Rots	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Collegerama	Yes	

CTB2200	Kansrekening & Statistiek	3
Responsible Instructor	Dr. C.O. Smet	
Co-responsible for assignments	Dr.ir. N.P.K. Chan	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-220-24	
Contact for students	Dr.ir. M.W. Ertsen

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-226-24	
Responsible Program	Prof.dr. A.P. Siebesma
Employee	
Minor Coordinator	Prof.dr. A.P. Siebesma

CT3080	The Science of Climate Change	5
Responsible Instructor	Dr. F. Glassmeier	
Instructor	M. Messmer	
Assistant	Dipl.-Ing. P. Alinaghi	
Contact Hours / Week x/x/x/x	2x2	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course will provide answers to the following questions: How is the climate changing? Are human activities responsible for global warming? What future changes can we expect? What do we know for sure and what is the most uncertain? Topics: 1. Introduction: Overview of the Climate System: atmosphere, ocean, cryosphere, land surface; energy balance; Greenhouse effect; radiative forcing, climate change. 2. Physics of the Climate System: - Fundamentals of the Atmosphere: chemical composition; compressibility and air pressure, thermal vertical stratification, global circulation. - Fundamentals of the Ocean & Cryosphere: properties of water; oceans layered structure; oceans global circulation; hydrological cycle; atmosphere ocean interactions. 3. The Carbon Cycle: - Reservoirs of carbon; the carbon cycle; acidification of the oceans; uncertainties in the carbon cycle 4. Climate System in Equilibrium - Global energy balance models of the Earth; consequences for the thermal structure of the atmosphere; radiative-convective equilibrium; geographical and seasonal variations in energy balance. 5. Climate Perturbations - Radiative forcing; natural vs anthropogenic forcings; main drivers of radiative forcing (CO₂, other Greenhouse gases, aerosols, land use) 6. Climate Feedbacks - Principles of feedback mechanisms; main feedbacks in our climate feedback System: water vapor, clouds, albedo; tipping points. 7. Paleoclimate - Milankovitch theory and ice age; five million years of climate; the last one hundred thousand years of climate change. 8. Climate Change in the Anthropocene: - A century of observed warming; natural vs anthropogenic drivers; precipitation and droughts; storms, hurricanes and other extremes - Sea-level rise; Greenland ice sheet; antarctic ice sheets; future sealevel rise. 9. Future Climate Model Predictions, - What are climate models; history of climate modeling; what can and what cannot be predicate by climate models; are climate models are only tool for looking into the future. 10. Design of Climate Scenarios - Climateemission scenarios; uncertainty in emission scenarios; properties of climate change risk; from global to regional scenarios.	
Study Goals	General Learning Goal: Critical understanding of the causes of climate change, the expected evolution of the climate, and the major uncertainties in current climate predictions. Learning Objectives: At the end of the course the student is able to: LO1: Give a basic description of the natural climate system. LO2: To explain, quantify and distinguish the basic natural and human induced forcings, and responses and feedbacks of the climate system. LO3: To explain the consequences of climate change for weather and weather extremes.	

	LO4: To identify and quantify the key uncertainties in climate predictions.
	LO5: To actively participate in a scientific debate on climate change.
Education Method	Alternating lecture and working classes. Most of the material will be put in practice for training and further exploration through the use of python notebook assignments.
Assessment	Mandatory set of assignments. Closed book exam, focus on analysis, interpretation and problem solving.
Expected prior Knowledge	Advanced highschool level of mathematics and physics; simple differential equations; elementary data analysis.
Academic Skills	Problem solving Thinking (critical, analytical) Interpretation Analysis Reasoning/arguing
Literature & Study Materials	Mathez and Smerdon: "Climate Change: The Science of Global Warming and our Energy Future" This book offers a mostly qualitative description of the course material. Participants are expected to read corresponding chapters before attending the lecture.
Judgement	Mandatory set of assignments. Closed book exam, focus on analysis, interpretation and problem solving. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Pocket calculator, provided list with formulas, constants, and definitions.
Collegerama	No

CT3080 Toets 1	<i>Final exam</i>	3.5
Responsible Instructor	Dr. F. Glassmeier	
Instructor	M. Messmer	
Assistent	Dipl.-Ing. P. Alinaghi	
Contact Hours / Week x/x/x/x	2x2	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See course description: CT3080	
Study Goals	See course description: CT3080	
Education Method	See course description: CT3080	
Assessment	See course description: CT3080	
Expected prior Knowledge	See course description: CT3080	
Academic Skills	See course description: CT3080	
Literature & Study Materials	See course description: CT3080	
Judgement	See course description: CT3080	
Permitted Materials during Exam	See course description: CT3080	
Collegerama	No	

CT3080 Toets 2	Group assignment	1.5
Responsible Instructor	Dr. F. Glassmeier	
Instructor	M. Messmer	
Assistent	Dipl.-Ing. P. Alinaghi	
Contact Hours / Week x/x/x/x	2x2	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See course description: CT3080	
Study Goals	See course description: CT3080	
Education Method	See course description: CT3080	
Assessment	See course description: CT3080	
Expected prior Knowledge	See course description: CT3080	
Academic Skills	See course description: CT3080	
Literature & Study Materials	See course description: CT3080	
Judgement	See course description: CT3080	
Permitted Materials during Exam	See course description: CT3080	
Collegerama	No	

CT3081-24	Climate Change Impact and Adaptation	5
Responsible Instructor	E. Ragno	
Course Coordinator	E. Ragno	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course will provide expert background on the impacts of climate change and on adaptation measures being enacted or considered around the world. An in-depth focus will be provided on freshwater resources, drought, coastal regions, and cities.	
Study Goals	The goals of the course are to understand the impacts of climate change on humans and the environment and to understand the effectiveness and limitations of adaptation options being considered. At the end of the course, students will be able to: (LO1) describe the impacts and adaptation of climate change on different sectors identified by the IPCC; (LO2) Explain the effect of climate change on: freshwater resources; coastal regions (low lying areas); the occurrence of droughts and heatwaves; (LO3) Identify potential adaptation strategies to: ensure freshwater availability, and allow life in coastal regions and under new climate conditions; (LO4) design adaptation measures for a case study site.	
Education Method	Lectures and flipped classroom.	
Assessment	Written exam (90% of the final grade) and project (10% of the final grade). Both the written exam and the project must be at least 5.8.	
Expected prior Knowledge	High school level mathematics and physics	
Academic Skills	Thinking (critical, analytical); Interpretation; Cooperation; Reasoning/arguing; Awareness of and reflection on and responsibility towards the social (international) context and consequences of technology and scientific actions	
Literature & Study Materials	lecture slides; IPCC reports; news article; journal papers; chapters from Resilience The science of Adaptation to Climate Change. All the necessary material will be available on the Brightspace page.	
Judgement	Written exam (90% of the final grade) and project (10% of the final grade). Both the written exam and the project must be at least 5.8.	
Permitted Materials during Exam	Pocket calculator	
Collegerama	No	

CT3081-24 Toets 1	Exam	4.5
Responsible Instructor	E. Ragno	
Course Coordinator	E. Ragno	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See Course Description: CT3081-24	
Study Goals	See Course Description: CT3081-24	
Education Method	See Course Description: CT3081-24	
Assessment	See Course Description: CT3081-24	
Expected prior Knowledge	See Course Description: CT3081-24	
Academic Skills	See Course Description: CT3081-24	
Literature & Study Materials	See Course Description: CT3081-24	
Judgement	See Course Description: CT3081-24	
Permitted Materials during Exam	See Course Description: CT3081-24	
Collegerama	No	

CT3081-24 Toets 2	Assignment	.5
Responsible Instructor	E. Ragno	
Course Coordinator	E. Ragno	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See Course Description: CT3081-24	
Study Goals	See Course Description: CT3081-24	
Education Method	See Course Description: CT3081-24	
Assessment	See Course Description: CT3081-24	
Expected prior Knowledge	See Course Description: CT3081-24	
Academic Skills	See Course Description: CT3081-24	
Literature & Study Materials	See Course Description: CT3081-24	
Judgement	See Course Description: CT3081-24	
Permitted Materials during Exam	See Course Description: CT3081-24	
Collegerama	No	

CT3082	Mitigation of Climate Change	5
Responsible Instructor	Dr.ir. S.E. Tanzer	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides an introduction to climate change mitigation and low carbon development. Climate change mitigation is about reducing greenhouse gas emissions to limit average global temperature increase. The largest contributor to these emissions is the energy system, including both energy supply and demand sectors, so a large part of this course focuses on energy. The course provides an overview of historic emissions and potential future emission pathways, covering technological options, economic feasibility and various policy instruments. The basics of energy systems, carbon accounting and emission mitigation analysis are explained, including interpretation of energy balances, emission reduction potentials and marginal abatement cost curves. Attention is paid to mitigation of both direct and indirect supply chain emissions in five major sectors: energy supply, buildings, transport, industry and agriculture and forestry.	
Study Goals	As a student, you will become familiar with the various options to mitigate climate change, know how to characterize them and understand how they interact. At the end of this course, you will be able to: - Have a multi-disciplinary understanding of climate change mitigation - Explain the basic principles of energy and carbon accounting - Describe, and perform calculations for, mitigation in different sectors - Perform simple technical, environmental and economic mitigation analyses - Evaluate different climate change mitigation policies and emission reduction potentials - Reflect on past, current and future trends in global emissions and emission abatement - Discuss the changes required for a transition to a low carbon future Climate change is a field where good argumentation skills are essential, so in addition to calculations, there is a large focus on critical thinking, discussion and reflection.	
Education Method	Lectures with room for exercises and group discussion. Further exercises are essential and tutorial classes are integrated in the to provide feedback and help with interpretation. Assignment is intended to provide an integrated perspective on mitigation.	
Assessment	Written exam, group assignment Judgement: written exam 70%, group assignment 30%, minimum grade of 5 required for each. Minimum participation of 80% in the lectures.	
Expected prior Knowledge	High school level mathematics and physics, basics in thermodynamics.	
Academic Skills	Problem solving Thinking (critical, analytical) Interpretation Reasoning/arguing Awareness of and reflection on and responsibility towards the social context and consequences of technology	
Literature & Study Materials	Textbook: K. Blok, E. Nieuwlaar, Introduction to Energy Analysis, Routledge, 3rd edition, 2021. Selected articles and excerpts from IPCC reports (Working Group III). Powerpoints of each lecture to be posted on Brightspace or Teams.	
Judgement	written exam 70%, group assignment 30%, minimum grade of 5 required for each. Minimum participation of 80% in the lectures. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	pocket calculator, text book and other course material	
Collegerama	No	

CT3082 Toets 1	Final exam	3.5
Responsible Instructor	Dr.ir. S.E. Tanzer	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See Course Description: CT3082	
Study Goals	See Course Description: CT3082	
Education Method	See Course Description: CT3082	
Assessment	See Course Description: CT3082	
Expected prior Knowledge	See Course Description: CT3082	
Academic Skills	See Course Description: CT3082	
Literature & Study Materials	See Course Description: CT3082	
Judgement	See Course Description: CT3082	
Permitted Materials during Exam	See Course Description: CT3082	
Collegerama	No	

CT3082 Toets 2	Group assignment	1.5
Responsible Instructor	Dr.ir. S.E. Tanzer	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See Course Description: CT3082	
Study Goals	See Course Description: CT3082	
Education Method	See Course Description: CT3082	
Assessment	See Course Description: CT3082	
Expected prior Knowledge	See Course Description: CT3082	
Academic Skills	See Course Description: CT3082	
Literature & Study Materials	See Course Description: CT3082	
Judgement	See Course Description: CT3082	
Permitted Materials during Exam	See Course Description: CT3082	
Collegerama	No	

CT3083-24	Climate Engineering	4
Responsible Instructor	I.M. Steinke	
Course Coordinator	I.M. Steinke	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to climate engineering (also called climate interventions). The two main principles, radiation management and carbon dioxide removal, will be discussed, as well as the proposed technologies, such as stratospheric aerosol injection, marine cloud brightening, biochar, afforestation and ocean fertilization. Attention will be given to the impact of the climate engineering technologies on the global and local living environment. We will also discuss ethical dilemmas and governance structures that are needed.	
Study Goals	To be familiar with and be able to assess the various options of climate engineering, ethics and governance. At the end of this course, you will be able to LO1: Explain the basic principles of different radiation management and carbon dioxide removal methods LO2: Describe the impact of climate engineering on the living environment LO3: Reflect on ethical dilemmas due to climate engineering LO4: Discuss possible governance structures Climate change is a field where good argumentation skills are essential, so in addition to calculations, there is a large focus on critical thinking, discussion and reflection.	
Education Method	Lectures with room for exercises and group discussion. Further exercises are essential and tutorial classes are organized to provide feedback and help with interpretation. Assignments are intended to provide an integrated perspective.	
Assessment	Written open book exam (e.g., essay writing), focus on analysis and interpretation. Group assignments, focus on integration perspectives	
Expected prior Knowledge	Highschool level of mathematics and physics	
Academic Skills	Problem solving Thinking (critical, analytical) Interpretation Reasoning/arguing Awareness of and reflection on and responsibility towards the social context and consequences of technology	
Literature & Study Materials	Selected articles and excerpts from IPCC reports (Working Group I-III) Scientific articles Powerpoints of each lecture to be posted on Brightspace	
Judgement	Group assignments (70%), individual assignment (30%), minimum grade: 5.8; active participation in class (pass/fail)	
Permitted Materials during Exam	Pocket calculator, course material.	
Collegerama	No	

CT3083-24 Toets 1	Group Assignment	2.8
Responsible Instructor	I.M. Steinke	
Course Coordinator	I.M. Steinke	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See course description: CT3083-24	
Study Goals	See course description: CT3083-24	
Education Method	See course description: CT3083-24	
Assessment	See course description: CT3083-24	
Expected prior Knowledge	See course description: CT3083-24	
Academic Skills	See course description: CT3083-24	
Literature & Study Materials	See course description: CT3083-24	
Judgement	See course description: CT3083-24	
Permitted Materials during Exam	See course description: CT3083-24	
Collegerama	No	

CT3083-24 Toets 2	Individual Assignment	1.2
Responsible Instructor	I.M. Steinke	
Course Coordinator	I.M. Steinke	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See course description: CT3083-24	
Study Goals	See course description: CT3083-24	
Education Method	See course description: CT3083-24	
Assessment	See course description: CT3083-24	
Expected prior Knowledge	See course description: CT3083-24	
Academic Skills	See course description: CT3083-24	
Literature & Study Materials	See course description: CT3083-24	
Judgement	See course description: CT3083-24	
Permitted Materials during Exam	See course description: CT3083-24	
Collegerama	No	

CT3084	Political Economy of Climate Change	5
Responsible Instructor	Dr. A.F. Correljé	
Instructor	R. Stöllinger	
Instructor	Dr.ir. R. van Bergem	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the political economy of climate change, considering the various consequences and options as regards mitigation and adaption from an economic perspective. What are the costs and benefits of climate change? Should we tax carbon, cap-and-trade emission rights, or subsidize solar and wind power? What are the arguments to choose one instrument, above the others? Why is it so hard to sign environmental agreements such as 'Kyoto' and 'Paris' despite the high stakes? Does trade harm the environment? Will our industry move abroad if we take a leading role and 'go it alone'? The course will consider these questions and the way to answers them, drawing on three different economic perspectives, namely the mainstream, the new institutional and the original institutional approach. Given the specific assumptions underlying these economic schools, they provide specific theories and methods of analysis, which are more or less relevant in answering the questions above. These assumptions involve key economic concepts, like actors rationality, uncertainty, efficiency, optimality, externalities, public goods, and property rights and most importantly the process of valuation and discounting. This allows us to discuss main issues, like the costs and benefits of climate change, regulatory principles of the UN Framework Convention on Climate Change, tradable permit system, clean development mechanism, joint implementation, etc. from these different perspectives.	
Study Goals	The module discusses the theory and practice of selecting the adequate forms of economic coordination in the understanding, design, implementation and execution of policies for climate change abatement. The objectives of the module are to develop the ability: 1) to reflect on the most important economic aspects of climate change and adaptation and mitigation policies. 2) to provide insights on climate change from mainstream, new institutional and the original institutional economics, referring to the assumptions underlying these economic schools, the specific theories and the relevant methods of analysis. 3) to judge, referring to these differences, what specific climate change issues should be analysed and understood with which economics approaches and concepts. 4) to identify, describe and interpret these issues, applying the paradigms and concepts of the different schools. 5) to critically reflect on public policy and firms strategies in climate change adaption and mitigation. 6) to identify the nature of the problems in public policy and firms in climate change adaption and mitigation, considering societal and technological developments. 7) To communicate about such issues, applying the appropriate economic terminology.	
Education Method	Lectures with room for group discussion. Exercises in using and applying the concepts in the course are essential and tutorial classes are organized to provide feedback and help with interpretation. A serious game is played in apply such insights in a 'live and interactive' setting.	
Assessment	Digital, open book exam, focus on analysis and interpretation. Participation in the serious game.	
Expected prior Knowledge	Basics in economics is advised; catch up material available on www.	
Academic Skills	The course draws on three different economic perspectives, namely the mainstream, the new institutional and the original institutional approach. Given the specific assumptions underlying these economic schools, they provide specific theories and methods of analysis, which are more or less relevant in answering different types questions in respect of climate change. The course thereto provides insight and helps to select the 'right' economic perspective.	
Literature & Study Materials	To be provided in the syllabus and on BS	
Judgement	Focus on understanding and ability to use concepts	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Course materials (USB or online)	
Collegerama	No	

CT3084 Toets 1	<i>Final exam</i>	3.5
Responsible Instructor	Dr. A.F. Correljé	
Instructor	Dr.ir. R. van Bergem	
Contact Hours / Week x/x/x/x	2 HC, 2 S, PS	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See course description: CT3084	
Study Goals	See course description: CT3084	
Education Method	See course description: CT3084	
Assessment	See course description: CT3084	
Expected prior Knowledge	See course description: CT3084	
Academic Skills	See course description: CT3084	
Literature & Study Materials	See course description: CT3084	
Judgement	See course description: CT3084	
Permitted Materials during Exam	See course description: CT3084	
Collegerama	No	

CT3084 Toets 2	<i>Group assignment</i>	1.5
Responsible Instructor	Dr. A.F. Correljé	
Instructor	Dr.ir. R. van Bergem	
Contact Hours / Week x/x/x/x	2 HC, 2 S, PS	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See course description: CT3084	
Study Goals	See course description: CT3084	
Education Method	See course description: CT3084	
Assessment	See course description: CT3084	
Expected prior Knowledge	See course description: CT3084	
Academic Skills	See course description: CT3084	
Literature & Study Materials	See course description: CT3084	
Judgement	See course description: CT3084	
Permitted Materials during Exam	See course description: CT3084	
Collegerama	No	

CT3085	End-Project	6
Responsible Instructor	Prof.dr. A.P. Siebesma	
Instructor	Dr. A.F. Correljé	
Instructor	E. Ragno	
Instructor	I.M. Steinke	
Instructor	Dr. F. Glassmeier	
Contact Hours / Week x/x/x/x	2	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	During this end-project the students will set up their own projects. They are encouraged to define a topic or case study within the field of climate science, climate engineering, mitigation adaptation or governance, to analyse it and, if possible, formulate possible solutions. A central theme in the end-project will be the integral approach of climate change problems. Each project theme will have a climate science component (the cause of the problem), climate impact (the relevance of the problem) a mitigation or adaptation strategy) (the solution) and the economical and political design and coordination of the intended solution (the implementation and feasibility). In practice an end-project might concentrate on one or two of these aspects but they will always be presented in the context of the complete chain from problem to implementation of the solution.	
Study Goals	LO1: Student is able to critically analyse and evaluate mitigation and adaptation solutions . LO2: : Student is able to critically analyse and evaluate probabilities, impacts and risks of climate change events. LO3: Student is able to present his/her analysis and evaluation in a clear, well organized manner in the form of a written manuscript as well as in a oral presentation. LO4: Student works constructively in a team; shows a professional attitude, takes initiatives, and is able to manage time & work.	
Education Method	Students will work in groups of 4-5 students on a specific topic/case study. A coach is assigned to each group (from the lecturers and other scientific staff of the involved departments). There will be a kick-off meeting, baseline meeting a mid-term meeting and a final review. During this meetings the students will be coached on monitored on the progress of their project. Simultaneously students will also receive instructions on: time management, how to do an experiment, how to use a scientific logbook. Scientific writing and reporting, searching the literature. Short tutorials on these soft skills will be offered by the coaches during weekly meetings. The final review consists of presentations from all groups (open to public) and feedback from the team of lecturers, involved scientific staff, and students. A final meeting with the coach and one member of the team of lecturers (per group) will be held to provide feedback in the written report.	
Assessment	Final group presentation Final group report	
Expected prior Knowledge	Proficient background in the basics of climate science & impacts, adaptation & mitigation and political and economic aspects of climate change demonstrated by attending the minor courses in Q1 and Q2.	
Academic Skills	Problem solving; critical and analytical thinking, understanding, analyzing and interpretation of academic text; communicating effectively in debates, academic writing;	
Literature & Study Materials	n/a	
Judgement	Group report 70%, group presentation & discussion 30% For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	n/a	

CT3085 Toets 1	Group report	4
Responsible Instructor	Prof.dr. A.P. Siebesma	
Instructor	Prof.dr.ir. H.W.J. Russchenberg	
Instructor	Dr. A.F. Correljé	
Instructor	Prof.dr. K. Blok	
Instructor	E. Ragno	
Contact Hours / Week x/x/x/x	See course description: CT3085	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See course description: CT3085	
Study Goals	See course description: CT3085	
Education Method	See course description: CT3085	
Assessment	See course description: CT3085	
Expected prior Knowledge	See course description: CT3085	
Academic Skills	See course description: CT3085	
Literature & Study Materials	See course description: CT3085	
Judgement	See course description: CT3085	
Permitted Materials during Exam	See course description: CT3085	
Collegerama	No	

CT3085 Toets 2	Group Presentation & Discussion	2
Responsible Instructor	Prof.dr. A.P. Siebesma	
Instructor	Prof.dr.ir. H.W.J. Russchenberg	
Instructor	Dr. A.F. Correljé	
Instructor	Prof.dr. K. Blok	
Instructor	E. Ragno	
Contact Hours / Week x/x/x/x	See course description: CT3085	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See course description: CT3085	
Study Goals	See course description: CT3085	
Education Method	See course description: CT3085	
Assessment	See course description: CT3085	
Expected prior Knowledge	See course description: CT3085	
Academic Skills	See course description: CT3085	
Literature & Study Materials	See course description: CT3085	
Judgement	See course description: CT3085	
Permitted Materials during Exam	See course description: CT3085	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

CT-MI-229-24

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Responsible for assignments	Dr.ir. A. Straub	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

CT3601	Exploratory Research Assignment	10
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Environmental engineering design process, in which lectures are given on different stages of the design process after which you should apply the knowledge in your own project. Cases will be where possible coupled to projects that are ongoing at the Green Village. You can thus use the inspiration you have obtained from this courses to develop circular solutions. After this stage, it is intended that the you select a few of the techniques and use them to come up with creative solutions for the larger societal challenges (sanitation, potable water supply, solid waste separation and recycling, energy).	
Study Goals	Students should challenge the end-of-pipe environmental engineering solutions and develop technological circular solutions in a structural method by which they follow the design phases that are common in the process engineering field. LO1: to be aware of methodology (TRIZ) that enhances creativity in exploring environmental engineering solutions LO2: to summarize and present the results of the preliminary design process and peerview the preliminary design process for other groups LO3: to organise the design process in several stages as typically applied in the environmental/process engineering domain and understand the cohesion between the different minor courses LO4: to apply theoretical knowledge of courses CT3602/ CT3603 /CT3604 /CTB2121/AR0097/ EE3105TU on a practical case LO5: to apply the LCA methodology as obtained in CT3602 on a practical case LO6: to create your own design (and prototype if possible) as a solution to a multidisciplinary environmental engineering challenge	
Education Method	Workshop on TRIZ problem-solving to engineering problems (short assignment with lectures) (28 hours). Groups of students will choose a case study and will be assigned 2 coaches (usually this will be the lecturers of the various courses). Regular meetings are planned in which progress is being reported. During these peer-2-peer feedback is given. A rubric will be formed to enable to judge the work on different topics systematically and equally.	
Assessment	Presentation of preliminary design in week 2 (summative) and final presentation end of Q2 (summative) Several progress presentations (formative) during Q2 Individual reflection (summative) on group cooperation, meeting of learning objectives, how they would approach the project if they had to redo it etc. Midterm group project report (formative) Final group project report (summative) Judgement: Rubrics will be made available during the course. Grades are divided over 10% preliminary design presentation, 10% final presentation, written report (70%), individual reflection (10%). Each group should provide peer feedback on the preliminary design and final presentation of another group. This is assessed as Pass / Fail	
Contact	r.e.f.lindeboom@tudelft.nl	
Expected prior Knowledge	Your expected to have at least attended all courses in Q1 of the minor EE&SD	
Academic Skills	Energy and mass balancing, development of block flow diagrams, process flow diagrams, P&ID and prototyping	
Literature & Study Materials	Articles & movies related to chosen topic Book: Triz for engineers, enabling inventive problem solving, Karin Gadd & Goddard Clive (accessible via TUDelft online library)	
Judgement	ubrics will be made available during the course. Grades are divided over 10% preliminary design presentation, 10% final presentation, written report (70%), individual reflection (10%). Each group should provide peer feedback on the preliminary design and final presentation of another group. This is assessed as Pass / Fail For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.	
Permitted Materials during Exam	N/A	
Collegerama	No	

CT3601 Toets 1	Voorontwerp Q1	1
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	See course description: CT3601	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	See course description: CT3601	
Study Goals	See course description: CT3601	
Education Method	See course description: CT3601	
Assessment	See course description: CT3601	
Expected prior Knowledge	See course description: CT3601	
Academic Skills	See course description: CT3601	
Literature & Study Materials	See course description: CT3601	
Judgement	See course description: CT3601	
Permitted Materials during Exam	See course description: CT3601	
Collegerama	No	

CT3601 Toets 2	Groepsrapportage	7
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	See course description: CT3601	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	See course description: CT3601	
Study Goals	See course description: CT3601	
Education Method	See course description: CT3601	
Assessment	See course description: CT3601	
Expected prior Knowledge	See course description: CT3601	
Academic Skills	See course description: CT3601	
Literature & Study Materials	See course description: CT3601	
Judgement	See course description: CT3601	
Permitted Materials during Exam	See course description: CT3601	
Collegerama	No	

CT3601 Toets 3	Eindpresentatie	1
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	See course description: CT3601	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	See course description: CT3601	
Study Goals	See course description: CT3601	
Education Method	See course description: CT3601	
Assessment	See course description: CT3601	
Expected prior Knowledge	See course description: CT3601	
Academic Skills	See course description: CT3601	
Literature & Study Materials	See course description: CT3601	
Judgement	See course description: CT3601	
Permitted Materials during Exam	See course description: CT3601	
Collegerama	No	

CT3601 Toets 4	Reflectieverslag	1
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	See course description: CT3601	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	See course description: CT3601	
Study Goals	See course description: CT3601	
Education Method	See course description: CT3601	
Assessment	See course description: CT3601	
Expected prior Knowledge	See course description: CT3601	
Academic Skills	See course description: CT3601	
Literature & Study Materials	See course description: CT3601	
Judgement	See course description: CT3601	
Permitted Materials during Exam	See course description: CT3601	
Collegerama	No	

CT3601 Toets 5	Peer assessment	0
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	See course description: CT3601	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	See course description: CT3601	
Study Goals	See course description: CT3601	
Education Method	See course description: CT3601	
Assessment	See course description: CT3601	
Expected prior Knowledge	See course description: CT3601	
Academic Skills	See course description: CT3601	
Literature & Study Materials	See course description: CT3601	
Judgement	See course description: CT3601	
Permitted Materials during Exam	See course description: CT3601	
Collegerama	No	

CT3602	Analysis of the Energy, Materials and Water Cycle in the Linear and Circular Economy	5
Responsible Instructor	Prof.dr. H.M. Jonkers	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	To properly understand the transition from a linear to a circular economy, it is necessary that the current linear model in which raw materials are used and discarded is well understood. LCA and energy analysis are useful tools that have been developed to map the sustainability of different processes in society. The analysis will show that throwing away products makes it virtually impossible to set up a sustainable society. Just like in nature, we will therefore have to extract all useful elements from waste products in order to be sustainable, for example by directly reusing the entire product or recycling components. This means that we must work towards a circular economic model in which waste must be valued as a new raw material. To achieve this transition, interventions must be made to reuse the substances. A case in form of an autonomous system will be worked out in which both required amounts of water, materials and energy and environmental performance of the system will be quantified.	
Study Goals	LO1 Applying Life Cycle Assessment (LCA) techniques for quantification of environmental impact of utility structures and civil engineering works (infra- and hydraulic structures) LO2 Quantification of environmental impact of renewable materials and energies LO3 Identification and characterization of circular materials LO4 Design of low environmental impact utility constructions and civil engineering works LO5 Identify bottlenecks and find solutions for successful market implementation of innovative circular- and low environmental impact materials	
Education Method	Lectures and workshops related to the 5 defined learning objectives of the course: 1. LCA techniques, 2. Renewable materials and energies, 3. Circular materials, 4. Integral design of multi-functional low environmental impact constructions and 5. Bottlenecks for successful market implementation of circular construction materials and products.	
Assessment	Case study portfolio grade contributes 30% and the written examination 70% to the final course grade. Minimum final grade must be 5,75 in order to pass the course.	
Expected prior Knowledge	None	
Academic Skills	Students learn how to quantify the environmental impact of building materials, processes and utility buildings and civil engineering works over their entire life cycle including use of water and energy. Quantification of environmental impact is done using Life Cycle Assessment (LCA) methods.	
Literature & Study Materials	Provided scientific publications and lectures on LO1-5 subjects	
Judgement	Case study portfolio grade contributes 30% and the written examination 70% to the final course grade. Minimum final grade must be 5,75 in order to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/ .	
Permitted Materials during Exam	Dictionary, calculator	
Collegerama	No	

CT3602 Toets 1	Exam	2.5
Responsible Instructor	Prof.dr. H.M. Jonkers	
Contact Hours / Week x/x/x/x	See Course Description: CT3602	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	See Course Description: CT3602	
Study Goals	See Course Description: CT3602	
Education Method	See Course Description: CT3602	
Assessment	See Course Description: CT3602	
Expected prior Knowledge	See Course Description: CT3602	
Academic Skills	See Course Description: CT3602	
Literature & Study Materials	See Course Description: CT3602	
Judgement	See Course Description: CT3602	
Permitted Materials during Exam	See Course Description: CT3602	
Collegerama	No	

CT3602 Toets 2	Report	2.5
Responsible Instructor	Prof.dr. H.M. Jonkers	
Contact Hours / Week x/x/x/x	See Course Description: CT3602	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	See Course Description: CT3602	
Study Goals	See Course Description: CT3602	
Education Method	See Course Description: CT3602	
Assessment	See Course Description: CT3602	
Expected prior Knowledge	See Course Description: CT3602	
Academic Skills	See Course Description: CT3602	
Literature & Study Materials	See Course Description: CT3602	
Judgement	See Course Description: CT3602	
Permitted Materials during Exam	See Course Description: CT3602	
Collegerama	No	

CT3603-23	Solid Waste to Secondary Material Resources	5
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Concepts in solid waste recycling and waste management, properties of waste sources and streams, materials characterization, sampling principles, waste analysis methodology, mechanical and sensor-based processing and sorting techniques, secondary resources and markets, circularity of materials recovered from solid waste.	
Study Goals	After the course the student should be able to: (1) recognize challenges in solid waste management; (2) recognize the potential and limitations of recycling routes; (3) have basic knowledge of modern solid waste processing techniques; (4) discuss aspects of solid waste recycling with peers; (5) place acquired knowledge in the context of circular economy.	
Education Method	Lectures and computer practicum (online in case of COVID restrictions)	
Assessment	Summative computer report on operating a waste recycling plant (1 EC) and final exam on the theory (4 EC). Minimum grade for both parts is 5.75	
Expected prior Knowledge	Admitted to the minor Environmental Engineering and Sustainable Design	
Academic Skills	knowledge of waste management and recycling techniques, able to put a specific waste problem in the context of circular economy	
Literature & Study Materials	lecture sheets and additional reading will be made available on Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE):	
Permitted Materials during Exam	basic calculator and pencils	
Collegerama	No	

CT3603-23 Toets 1	Computer practicum	1
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	See Course Description: CT3603-23	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See Course Description: CT3603-23	
Study Goals	See Course Description: CT3603-23	
Education Method	See Course Description: CT3603-23	
Assessment	See Course Description: CT3603-23	
Expected prior Knowledge	See Course Description: CT3603-23	
Academic Skills	See Course Description: CT3603-23	
Literature & Study Materials	See Course Description: CT3603-23	
Judgement	See Course Description: CT3603-23	
Permitted Materials during Exam	See Course Description: CT3603-23	
Collegerama	No	

CT3603-23 Toets 2	Tentamen	4
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Contact Hours / Week x/x/x/x	See Course Description: CT3603-23	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See Course Description: CT3603-23	
Study Goals	See Course Description: CT3603-23	
Education Method	See Course Description: CT3603-23	
Assessment	See Course Description: CT3603-23	
Expected prior Knowledge	See Course Description: CT3603-23	
Academic Skills	See Course Description: CT3603-23	
Literature & Study Materials	See Course Description: CT3603-23	
Judgement	See Course Description: CT3603-23	
Permitted Materials during Exam	See Course Description: CT3603-23	
Collegerama	No	

CT3604	Biomass Nutrients and Energy Conversion	2
Responsible Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Prof.dr. H.M. Jonkers	
Instructor	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Biomass is used as a source of nutrients (compost) and renewable energy. Specifically, biomass is responsible for 61% of green energy production in the Netherlands (2019), but the total contribution of green energy is only 7.4%. That is why biomass for energy is subsidised to stimulate rapid growth in order to meet the European climate and energy ambitions for 2030. But large scale application of biomass also invokes controversy, such as the facts surrounding incineration, conflict with food production and reduced biodiversity, just to name a few. This course aims to give a wide view on biomass, ranging from societal and legislative drives, availability and potential, technologies, energy conversion processes, as well as applications of biomass and their impacts on the environment.	
Study Goals	After the course the student should be able to: (1) have basic knowledge of biomass as a nutrient and as energy source; (2) calculate basic physical and chemical properties of biomass flows and energy conversion rates; (3) discuss biomass challenges with peers; (4) recognise the environmental and climate related implications of biomass.	
Education Method	Lectures	
Assessment	Written exam. Minimum grade is 5.75	
Expected prior Knowledge	entrance to minor Environmental Engineering and Sustainable Design	
Academic Skills	basic control of chemistry, energy concepts and mathematics.	
Literature & Study Materials	Lecture sheets and additional reading will be made available on Brightspace	
Judgement	.	
Permitted Materials during Exam	basic calculator and pencil	
Collegerama	No	

CTB2121	Urban Water and Environmental Technology	5
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Prof.dr.ir. J.B. van Lier	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

CTB2121 Toets 1	Deeltentamen Stedelijk Water	0
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Prof.dr.ir. J.B. van Lier	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

CTB2121 Toets 2	Deeltentamen Milieutechniek	0
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Prof.dr.ir. J.B. van Lier	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

CTB2121 Toets 3	<i>Integraal tentamen</i>	3
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Prof.dr.ir. J.B. van Lier	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

CTB2121 Toets 4	<i>Integrale opdracht</i>	2
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

CTB2121 Toets 5	<i>Workshop</i>	0
Responsible Instructor	Prof.dr.ir. L.C. Rietveld	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Collegerama	No	

EE3105TU	Sustainable Energy Technologies	3
Responsible Instructor	Dr.ir. H. Ziar	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	A transition to sustainable energy is needed for our climate and welfare. This course is an introduction to the Master Programme Sustainable Energy Technology at TU Delft and is aimed at Bachelor students from science and engineering disciplines. In this course, you will learn how to assess the potential for energy-use reduction and the potential of generating and storing renewable energy sources such as wind, solar and hydro. You'll learn how the flow of energy from various sources interact with different consumption sectors and how power plants participate in electricity markets. Course content: 1. Climate change and role of sustainable energies 2. Energy access and consumption of energy 3. Generation of renewable energy: Hydro-power, wind, and solar PV 4. Energy Balances 5. Electricity Markets 6. Storage of energy	
Study Goals	Learning objectives: LO1: Describe the reasoning behind global warming and link to the need for sustainable energy technologies. LO2: Assess resources potential, energy use, and ways to reduce consumption at different sectors (transport, industry, buildings) LO3: Calculate electricity generation from various renewable energy plants, such as wind, solar, and hydro LO4: Analyze energy balance sheets, link them with Sankey diagrams and assess future energy policies LO5: Rank merit order of generation and calculate power plant earnings in electricity market LO6: Calculate the capacity of various energy storage technologies, such as flywheel, pump-hydro, compressed air, battery, and super-capacitors	
Education Method	Lectures/instructions, video lectures, Exercises	
Assessment	The course EE3105TU Sustainable Energy Technologies will be assessed in Q1 and in Q2 (resit) of the academic year as follows: The exam is a written on-campus exam. The exam consists of open questions and multiple-choice questions. The exam is a closed book but you can bring one page of formula sheet. Like the conventional on-campus exam, you can use only pen, paper and a calculator. The exam takes 3 hours. Communication with others is not allowed! The exam will be assessed with a mark with a resolution of 0.5. A score equal to or higher than 58.0 out of 100 is needed to pass the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Economics Energy Energy & Industry Industry Physics Sustainability Technology	

Year

Organization

Education

2024/2025

Civil Engineering and Geosciences

Minors Civil Engineering

CT-MI-234-24	
Responsible Program Employee	Dr.ir. A.J. Pel
In association with the Faculty of	CITG
Administration by the Faculty of	CITG
Minor Title	Mobility & Transport (M&T)

CT3501	Mobiliteit en Bereikbaarheid	4
Responsible Instructor	Dr. C. Maat	
Co-responsible for assignments	Dr.ir. A.J. van Binsbergen	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Collegerama	No	

CT3502	Het Transportsysteem	4
Responsible Instructor	Dr.ir. N. van Oort	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Education Method	On campus colleges over selectie aan onderwerpen.	
Assessment	Schriftelijk tentamen over volledig aanbod aan onderwerpen.	
Expected prior Knowledge	CT3500 Introductieproject: kennis van lagenmodel.	
Literature & Study Materials	John Baggen & Hans van Ham, Het Transportsysteem. Delft Academic Press, 2019. ISBN 9789065624406. Vd Bijl & Van Oort, Betere Bus, Acquire publishing Beschikbaar via secretariaat Transport & Planning (eind Aug)	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Gesloten boektentamen	
Collegerama	No	

CT3503	Stad en Transport	4
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	The course provides an introduction to the field of work of urban and transport planners. Topics covered are: Urban Development - Historical urban development in relation with the development of transport systems and networks; - Analysis and Design techniques for urban areas including Noli, Morphological, Lynch, and Cullen analysis (qualitative). Transport systems - Infrastructure, means of transport (vehicles) and services; - Relationships between transport demand, transport services and the traffic system; - Analysis and modeling techniques, including 4-step model, microsimulation, (system) dynamic analysis (qualitative); - Characteristics of transport and traffic networks: network elements, hierarchy in networks, scaling, combining and integration of subsystems. Urban development and transportation systems - Relationships between area characteristics, transport demand and transport and traffic; - Relationship between urban development and infrastructure: infrastructure as the guiding principle for spatial design as well as the transport consequences of spatial planning; the inertia of infrastructure; - Different perspectives on urban development regarding transport systems and concepts, including: Grids, Garden Cities, Transit Oriented Development, 15 Minutes City (including attention to 'active modes'). Practical examples: restructuring of former port areas, public transport and urban development, inner-city development. Analyzing and designing an integrated urban spatial plan and traffic system in one large assignment.	
Study Goals	LO1 be able to explain the backgrounds of the form and function of cities, be able to explain what important urban functions are and how they are related to traffic and transport, be able to identify principles for analyzing and designing urban and traffic systems; LO2 to be able to explain the relationship between urban activities and transport and traffic by means of conceptual models and qualitative relationships; LO3 analyzing and comparing urban and traffic structures using analysis techniques; LO4 to evaluate and to qualitatively assess the quality of urban areas and associated traffic systems; LO5 to develop a program of requirements and to design an integrated spatial and traffic plan for an urban area to be re- or newly developed.	
Education Method	Lectures and seminars (with guest speakers), workshops and projectwork (both as groupwork), centered around a case study; optional (and self-organised) site-visit/fieldwork The lectures will be supported by documents and short movies on specific topics available on BrightSpace. A site-visit to the study case location is strongly encouraged; this site-visit is to be organised by students itself.	
Assessment	Grading of Group assignment and an additional Individual Exam (digital, written). Both the Group Assignment and the Individual Exam will be graded.	
Remarks	NOTE: spoken language is Dutch, course material (slides) is in English	
Expected prior Knowledge	CT3501 - Het Ruimtelijk Systeem: Mobiliteit en Effecten en CT3502 - Het Transportsysteem: Netwerken en Mobiliteit	
Academic Skills	Systematic Analysis, Use of scientific sources, Systematic design, Knowledge integration	
Literature & Study Materials	Lecture slides; advised, but not mandatory: Carmona, M. et al. Public Places, Urban Spaces, The Dimensions of Urban Design, 2nd edition, Elsevier Architectural Press, Oxford 2010	
Judgement	Individual Exam: weight 60%, minimum partial grade: 5.0 Group Assignment (report): weight 40%, minimum partial grade: 6.0 Overall pass grade required: 5.8	
Permitted Materials during Exam	Calculator (non-programmable)	
Collegerama	No	

CT3505-24	Traffic Flows and Management	4
Responsible Instructor	Dr.ir. M. Rinaldi	
Course Coordinator	Dr.ir. M. Rinaldi	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Familiarity with transportation demand and supply, notions covered in the other courses of the M&T minor. Experience with programming, particularly in Python and Jupyter Notebooks is recommended. In preparation for this course, the following teach book is recommended for all students: https://teachbooks.github.io/learn-python/intro.html	
Course Contents	This course aims to introduce students to concepts, theories and models in the field of transportation systems. The course balances a qualitative and quantitative aspects of transportation science, and modelling. The base elements of transportation network supply are introduced, at different levels of aggregation (microscopic scale, macroscopic scale), with the related modelling principles. These different modelling scales are then associated to corresponding traffic flow phenomena, with consideration of the cause-effect chain arising between different aggregation levels. Multiple modes of transport are discussed through comparative analysis, considering what changes between different modes and what instead forms an underlying common ground. The course finally introduces aspects of transportation network design, discussing the relevance of objectives such as safety and emissions, the importance and practices related to data collection, and management techniques and how these address different objectives. Course content: Traffic operations (roads, intersections, vehicle interactions, flow phenomena) Measuring traffic: traffic data, sensing equipment and fundamental quantities, relation between microscopic dynamics and macroscopic dynamics Microscopic traffic flow dynamics and models: car-following and lane changing Macroscopic traffic flow dynamics and models: KWT, base quantities, shockwave theory and speeds, shockwave examples. Comparative analysis of other modes: active modes (pedestrians, bikes), mixed traffic flows (bus, trucks) and emerging mobility Traffic externalities & impact assessment: traffic safety and emissions Reducing externalities, governance: traffic control, network management	
Study Goals	At the end of this course you will be able to: - Identify the building blocks of a transportation system - Explain the fundamentals of traffic modeling and driving behaviour at different aggregation levels - Understand how these fundamentals vary for different and new modes of transport - Evaluate an existing traffic network and its capabilities and limitations - Analyze the requirements for creating an efficient traffic control system	
Education Method	The course is based on lectures, featuring oral presentations (including guest lectures) and computer based exercises (Python, Jupyter notebook).	
Computer Use	The course features Python and Jupyter notebooks, in combination with the microscopic traffic flow simulation software SUMO, in support of its first summative assessment report. Students are recommended to bring their own laptops for respective lectures, where the given notebooks are going to be introduced. Students are encouraged to familiarise themselves with conda environments prior to the course. See more here: https://docs.conda.io/en/latest/. A lightweight version of the environment is available here: https://docs.anaconda.com/free/miniconda/. SUMO can be downloaded and installed here: https://eclipse.dev/sumo/	
Course Relations	The course is part of the Mobiliteit en Transport minor of the Civil Engineering faculty, it is thereby complementary in content with other courses in the minor. The course features notions of Traffic Flow Theory and Traffic Management, as taught in the Transport and Traffic Engineering track of the MSc. degree in Civil Engineering, Track Base (TB3) and B2 modules.	
Literature and Study Materials	The course material and structure are (loosely) based on the following textbooks: "Victor Knoop - Traffic Flow Theory: An introduction with exercises" https://textbooks.open.tudelft.nl/textbooks/catalog/book/39 and "Martin Treiber, Arne Kesting (2013) - Traffic Flow Dynamics: Data, Models and Simulation" ISBN 978-3-642-32460-4. Lecture slides, Jupyter notebooks and course materials for students needing a fast Python programming primer are also available.	
Books	Victor Knoop (2021) - Traffic Flow Theory: An introduction with exercises - ISBN 978-94-6366-378-6. Martin Treiber, Arne Kesting (2013) - Traffic Flow Dynamics: Data, Models and Simulation - ISBN 978-3-642-32460-4.	
Assessment	Assessment is based on two summative assignments, carried out during the course itself. The first assignment (TOETS 1) is a group-based report based on simulation carried out with the help of Jupyter Notebook and the software SUMO. This report concerns aspects of Traffic Flow Theory and Management. The second assignment (TOETS 2) is a group-based report, where students identify an engineering challenge in transportation from their personal lives/experiences and propose a conceptual design for a potential solution. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Expected prior Knowledge	Familiarity with transportation demand and supply, notions covered in the other courses of the M&T minor. Experience with programming, particularly in Python and Jupyter Notebooks is recommended. In preparation for this course, the following teach book is recommended for all students: https://teachbooks.github.io/learn-python/intro.html	
Academic Skills	High school level calculus, calculus of variations and linear algebra.	
Literature & Study Materials	The course material and structure are (loosely) based on the following textbooks: "Victor Knoop - Traffic Flow Theory: An introduction with exercises" https://textbooks.open.tudelft.nl/textbooks/catalog/book/39 and "Martin Treiber, Arne Kesting (2013) - Traffic Flow Dynamics: Data, Models and Simulation" ISBN 978-3-642-32460-4. Lecture slides, Jupyter notebooks and course materials for students needing a fast Python programming primer are also available.	
Judgement	Grading rubrics for the two summative assignments are available on Brightspace.	
Permitted Materials during Exam	None	
Collegerama	Yes	

CT3505-24 Toets 1	<i>Traffic flow modelling and management - from dynamics to control policy</i>	2
Responsible Instructor	Dr.ir. M. Rinaldi	
Course Coordinator	Dr.ir. M. Rinaldi	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See course description: CT3505-24	
Study Goals	See course description: CT3505-24	
Education Method	See course description: CT3505-24	
Assessment	See course description: CT3505-24	
Expected prior Knowledge	See course description: CT3505-24	
Academic Skills	See course description: CT3505-24	
Literature & Study Materials	See course description: CT3505-24	
Judgement	See course description: CT3505-24	
Permitted Materials during Exam	See course description: CT3505-24	
Collegerama	No	

CT3505-24 Toets 2	<i>Mobility management system - conceptual model</i>	2
Responsible Instructor	Dr.ir. M. Rinaldi	
Course Coordinator	Dr.ir. M. Rinaldi	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See course description: CT3505-24	
Study Goals	See course description: CT3505-24	
Education Method	See course description: CT3505-24	
Assessment	See course description: CT3505-24	
Expected prior Knowledge	See course description: CT3505-24	
Academic Skills	See course description: CT3505-24	
Literature & Study Materials	See course description: CT3505-24	
Judgement	See course description: CT3505-24	
Permitted Materials during Exam	See course description: CT3505-24	
Collegerama	No	

CT3510	Onderzoeksproject Mobiliteit	7
Responsible Instructor	Dr. C. Maat	
Co-responsible for assignments	Dr.ir. A.J. Pel	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	Dutch	
Collegerama	No	

CT3520	Seminars in M&T: Onderzoeksvelden en Trends	3
Responsible Instructor	Dr.ir. A.J. Pel	
Contact Hours / Week x/x/x/x	1/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	The course consists of 2 components: (1) seminars and (2) trend analysis in which groups of 2-3 students work together. (1) Seminars A series of one or two seminars per week (in total ~10) present the main research fields of the domain of Mobility & Transport, such as rail traffic, electric and automated transport, traffic safety, public transport, freight and logistics, active modes, traffic control, smart and sustainable mobility, traffic simulation and data analytics. The aim of each seminar is to provide an overview of the current status of that field (i.e., scope, applications, stakeholders/partners), how it is being researched (i.e., projects, lab facilities, data, methods) and its future challenges (i.e., goals, developments). A seminar is approximately 3 hours consisting of presentations, demonstrations, discussions, etc. and is each time provided by one of the research labs from the Transport & Planning department at TU Delft. (2) Trend analysis As a group, a trend is chosen (in consultation with the course coordinator) and analysed. The trend should be external to the M&T domain, but related to at least one of the main research fields. Examples of such 'trends' could relate to densification of urban areas, online retail, remote working/learning, energy transition, air quality, climate change, natural disasters and emergencies, global economy, public health, aging populations, inclusiveness, digitalization of information and services, internet-of-things, AI, and more. You are highly encouraged to think of a trend from within your own BSc field that may interact with the M&T domain. The analysis includes: - examining the context, driving factors, goals and consequences of the trend, particularly those related to M&T. - identifying and describing the roles of research fields of M&T therein and how they (can) contribute. The trend analysis should be based on the understanding of the M&T domain and research fields as gained in the seminars, as well as on a search and review of the (scientific) literature. Seminar attendance: 10 x 3h = 30h Trend analysis: 54h Total: 84 hours (= 3 EC)	
Study Goals	In a series of seminars and group work students learn to understand main research fields of Mobility & Transport in terms of their scope, current knowledge, applications, research methods, lab facilities and future challenges, and how these fields interact with domain-specific developments as well as wider trends in society. After following this course, students should be able to: - identify and describe main research fields of M&T in terms of their scope, current status, research methods and challenges. - search and review (scientific) literature for M&T research fields. - analyse and report on a trend related to the M&T domain and the roles of multiple research fields of M&T therein.	
Education Method	Seminars, discussion, literature study, self-study, group work.	
Assessment	Students submit as a group before the end of week 10 a product that presents their trend analysis. The form of the product should be chosen appropriately. Examples are a poster, explainer video, slide show or report. Grading (group grade) is based on the quality of analysis of the trend (30%) and roles of research fields (30%), the literature study (20%) and the product (20%). A grading rubric will be provided. Pass/fail criterion additional to grading: Seminar attendance is mandatory. A maximum of 2 seminars can be missed, under the condition that any missed seminar must be compensated by self-study to fulfil the seminar learning goals and a report thereof.	
Expected prior Knowledge	None	
Academic Skills	Analytical thinking, technical and scientific reasoning, discussion, literature study, self-study, organizing, reporting.	
Literature & Study Materials	Seminars may include presentation material that will be provided.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CT3521	Seminars in M&T: Case Studies	4
Responsible Instructor	Dr.ir. A.J. Pel	
Contact Hours / Week x/x/x/x	weeks 1-6: 3 lecture hours per week (consultation hours) weeks 7-8: 2 days per week (seminars)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of 2 phases in which groups of 4-6 students work together to (1) analyse multiple case studies and (2) present their findings in a seminar. (1) Case study analysis, weeks 1-6 Groups choose (in consultation with the course coordinator) a type of infrastructure, technology, service or system that is related to the domain of M&T and has recently been implemented. Examples are mobility hubs, park-and-ride facilities, large-scale bicycle parking, in-car driver assistance technologies, connected vehicles, autonomous shuttles, long-distance bus services, sustainable city logistics, various forms of shared mobility, car-free inner cities, shared space, transit-oriented development, network-wide traffic control, smart traffic lights and road pricing. In weeks 1-6 groups search, select and analyse multiple associated case studies. The analysis includes examining the spatial, technological and societal aspects of these specific case studies, how they integrate in the local/regional multimodal transport system, and their performance and (sustainability) impacts. These insights should then be synthesized towards main lessons learned and evaluated in terms of theoretical support, empirical evidence and generalisability (where students can also draw from the understanding gained in the other Minor courses). (2) Seminars, weeks 7-9 In weeks 7-8 groups prepare a seminar to present their insights from the case study analysis phase. The preparation includes formulating learning goals, choosing effective seminar activities, setting the agenda, distributing roles/responsibilities to all group members to execute the seminar event and creating the (presentation) materials to be used at the seminar. Seminars are held in week 9. A seminar is approximately 45-60 minutes of presentations and audience engagement (e.g., discussions, polls, etc.) and is moderated by the group.	
Study Goals	In groups students learn to draw insights from across multiple case studies in the domain of Mobility & Transport in terms of spatial and socio-technological aspects, integration in the transport system and sustainability impacts, and to present these insights in a seminar for fellow students. After following this course, students should be able to: - identify case studies related to a type of infrastructure, technology, service or system in the domain of M&T - analyse case studies in terms of spatial and socio-technological aspects, integration in the transport system and sustainability impacts. - synthesize findings from multiple case studies and evaluate the main lessons learned in terms of theoretical support, empirical evidence and generalisability. - organise and moderate a seminar (for fellow students) in terms of learning goals, activities, agenda, presentation material and audience engagement.	
Education Method	Case studies, literature study, group work, seminar organisation, seminars	
Course Relations	This course is part of the Minor programme Mobiliteit & Transport	
Assessment	Students submit as a group by end of week 8: - an outline of their seminar (approximately 3 pages), including the learning goals, activities, agenda, and roles/responsibilities of all group members. - all (presentation) materials to be used at their seminar. Seminars are held in week 9. Grading: The seminar is graded (group grade) based on the quality of analysis of case studies (40%), quality of synthesis of main findings (40%), and organisation and moderation of the seminar event (20%). A grading rubric will be provided. Pass/fail criterion additional to grading: Seminar attendance is mandatory. A maximum of 2 seminars can be missed, under the condition that any missed seminar must be compensated by self-study to fulfil the seminar learning goals and a report thereof.	
Expected prior Knowledge	Previous minor M&T courses	
Academic Skills	Analytical thinking, technical and scientific reasoning, literature study, self-study, organizing, presenting.	
Literature & Study Materials	Introduction lecture with outline and tips. Weekly consultation hours	
Judgement	See assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Not applicable	
Colleggerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Minors Civil Engineering

Vrije minor extern project 10/15 EC	
Program Coordinator	Richard Beugeling

CTBAESB3010	Free minor project	10
Responsible Instructor	Dr.ir. M.W. Ertsen	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Collegerama	No	

CTBAESB3015	Free minor project	15
Responsible Instructor	Dr.ir. M.W. Ertsen	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Collegerama	No	

Dipl.-Ing. P. Alinaghi

Department	Atmospheric Remote Sensing
Department	Atmospheric Remote Sensing

E. van Andel

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89317
Room	31.C2.090

Dr. K. Anupam

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82394
Room	23.HG 6.60

A.W. Baar

Department	Surface and Groundwater Hydrology
-------------------	-----------------------------------

Dr.ir. M.C.M. Bakker

Unit	Civiele Techniek & Geowetensch
Department	Resources & Recycling
Telephone	+31 15 27 85219

Dr.ir. R. van Bergem

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 84819

Dr.ir. A.J. van Binsbergen

Unit	Civiele Techniek & Geowetensch
Department	Transport, Mobility and Logistics
Telephone	+31 15 27 81067
Room	23.HG 4.07

Prof.dr. K. Blok

Department	Energie en Industrie
Telephone	+31 15 27 89284
Room	31.B3.060
Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 89284
Room	31.B3.060

Drs.ing. J.J. de Boer

Unit	Universiteitsdienst
Department	Research Support & Innovation
Telephone	+31 15 27 84620
Room	23.HG 5.13

Dr.ir. J. Bosboom

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 84606
Room	23.HG 3.66

M.G.C. Bosch-Rekvelde

Unit	Civiele Techniek & Geowetensch
Department	Integral Design & Management
Telephone	+31 15 27 84771
Room	23.HG 6.69

Prof.mr.dr. J.A. de Bruijn

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 87169

Dr. A. Cabboi

Department	Mechanics and Physics of Structures
Telephone	+31 15 27 83419
Room	23.S2 1.54

Dr.ir. N.P.K. Chan

Department	Statistics
Room	36.HB 06.130

Dr.ir. A.M.J. Coenders

Unit	Civiele Techniek & Geowetensch
Department	Water Systems Monitoring & Modelling
Telephone	+31 15 27 82110
Room	23.HG 4.95

Dr. A.F. Correljé

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 87924

Dr.ir. A.M. Droste

Department	Water Systems Monitoring & Modelling
-------------------	--------------------------------------

Dr.ir. M.W. Ertsen

Unit	Civiele Techniek & Geowetensch
Department	Water Systems Engineering
Telephone	+31 15 27 87423
Room	23.HG 4.81

Dr.ir. A. Gavrilidou

Department	Traffic Systems Engineering
Room	23.HG 4.40

Ir. F. Geerts

Unit	Bouwkunde
Department	Theory, Territories & Transitions
Telephone	+31 15 27 83754
Room	08.01OOST700

Dr.ir. R. Gelderloos

Department	Environmental Fluid Mechanics
Telephone	+31 15 27 81942

Dr. F. Glassmeier

Department	Atmospheric Remote Sensing
Telephone	+31 15 27 83541
Room	23.HG 2.35

Prof.dr.ir. D. van Halem

Unit	Civiele Techniek & Geowetensch
Department	Sanitary Engineering
Telephone	+31 15 27 85838
Room	23.HG 4.48

Prof.dr.ir. M.J.C.M. Hertogh

Unit	Civiele Techniek & Geowetensch
Department	Integral Design & Management
Telephone	+31 15 27 84921
Room	23.HG 6.63

Dr. E.J. Houwing

Department	Integral Design & Management
Room	23.HG 6.71

Dr.ir. I.U. Hurkxkens

Department	Landschapsarchitectuur
Department	Landschapsarchitectuur

Dr.ir. R.W. Hut

Unit	Civiele Techniek & Geowetensch
Department	Water Resources
Telephone	+31 15 27 84909
Room	23.HG 4.76

Dr. L.C.M. Itard

Unit	Bouwkunde
Department	Environmental & Climate Design
Telephone	+31 15 27 86341
Room	08.01WEST540

Prof.dr. H.M. Jonkers

Unit	Civiele Techniek & Geowetensch
Department	Materials- Mechanics- Management & Design
Telephone	+31 15 27 82313
Room	23.HG 6.07

Ir. C. Kasbergen

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82729
Room	23.S2 2.36

S.J. Kestens

Unit	Civiele Techniek & Geowetensch
Department	Management Support
Telephone	+31 15 27 87515
Room	23.HG 4.52

T.T. Kettler

Department	Coastal Engineering
Room	23.HG 3.62

mr. J.M. Kooijman

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 81173
Room	31.b2.220

Dr.ir. P.A. Korswagen Eguren

Department	Applied Mechanics
Telephone	+31 15 27 84689

Department	Applied Mechanics
Telephone	+31 15 27 84689
Room	23.HG 6.57

Prof.dr.ir. M.K. de Kreuk

Unit	Civiele Techniek & Geowetensch
Department	Water Management
Telephone	+31 15 27 85274

Dr. Ir. R. Kuttantharappel Soman

Department	Integral Design & Management
Telephone	+31 15 27 85260

Dr. M. Leijten

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 83561

Prof.dr.ir. J.B. van Lier

Unit	Civiele Techniek & Geowetensch
Department	Sanitary Engineering
Telephone	+31 15 27 81615
Room	23.HG 4.57

Dr.ir. R.E.F. Lindeboom

Unit	Civiele Techniek & Geowetensch
Department	Sanitary Engineering
Telephone	+31 15 27 82069
Room	23.HG 4.61

Dr. C. Maat

Unit	Bouwkunde
Department	Transport, Mobility and Logistics
Telephone	+31 15 27 87640
Room	23.HG 4.25

Dr. A. Meijer

Unit	Bouwkunde
Department	Environmental & Climate Design
Telephone	+31 15 27 85658
Room	08.01+.West.020

M. Messmer

Department	Atmospheric Remote Sensing
-------------------	----------------------------

Ir. G.J.M. van der Meulen

Department	Urban Design
Telephone	+31 15 27 83299

Department	Urban Design
Telephone	+31 15 27 83299
Room	08.BG.West.160

Dr.ir. E. Minkman

Department	Organisatie en Governance
-------------------	---------------------------

Dr. E. Mostert

Unit	Civiele Techniek & Geowetensch
Department	Surface and Groundwater Hydrology
Telephone	+31 15 27 87800
Room	23.HG 4.82

J. Ninan

Department	Integral Design & Management
Telephone	+31 15 27 89608

Dr. A.A. Núñez Vicencio

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 89355
Room	23.S2 2.35

Dr.ir. N. van Oort

Unit	Civiele Techniek & Geowetensch
Department	Transport, Mobility and Logistics
Room	23.HG 4.10.2

Ir. M. van Os

Department	Integral Design & Management
-------------------	------------------------------

Prof.dr. M. Overend

Department	Structures & Materials
-------------------	------------------------

N.G. Ozerkan

Department	Materials and Environment
-------------------	---------------------------

Dr.ir. A.J. Pel

Unit	Civiele Techniek & Geowetensch
Department	Transport, Mobility and Logistics
Telephone	+31 15 27 84874
Room	23.HG 4.35

E. Ragno

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Room	23.HG 3.75

Prof.dr.ir. L.C. Rietveld

Unit	Civiele Techniek & Geowetensch
Department	Sanitary Engineering
Telephone	+31 15 27 84732
Room	23.HG 4.53

Dr.ir. M. Rinaldi

Department	Traffic Systems Engineering
-------------------	-----------------------------

Prof.dr.ir. J.G. Rots

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 83799
Room	23.HG 6.71

Ir. H. Rozendaal

Department	Structures & Materials
Room	08.01+.West.210

Prof.dr.ir. H.W.J. Russchenberg

Unit	Civiele Techniek & Geowetensch
Department	Atmospheric Remote Sensing
Telephone	+31 15 27 86292
Room	23.HG 2.10

N. Sanaan Bensi

Department	Theory, Territories & Transitions
Room	08.01.Oost.430

Dr.ir. M.A. de Schipper

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 87692
Room	23.HG 3.68

Dr. M.A. Schleiss

Unit	Civiele Techniek & Geowetensch
Department	Atmospheric Remote Sensing
Telephone	+31 15 27 89204
Room	23.HG 2.24

Dr.ir. G.H.W. Schoups

Unit	Civiele Techniek & Geowetensch
Department	Surface and Groundwater Hydrology
Telephone	+31 15 27 84508
Room	23.HG 4.94

Prof.dr. A.P. Siebesma

Unit	Civiele Techniek & Geowetensch
Department	Geoscience and Remote Sensing
Telephone	+31 15 27 85628
Room	23.KG 02.15

Dr. C.O. Smet

Department	Statistics
Telephone	+31 15 27 88666
Room	36.HB 06.260

Dr.ir. J.E.P. Smits

Unit	Bouwkunde
Department	Structures & Materials
Telephone	+31 15 27 88673
Room	08.01+.West.130

I.M. Steinke

Department	Atmospheric Remote Sensing
-------------------	----------------------------

Dr.ir. A. Straub

Unit	Bouwkunde
Department	Design & Construction Management
Telephone	+31 15 27 82769

R. Stöllinger

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 88583

Dr.ir. S.E. Tanzer

Department	Energie en Industrie
Telephone	+31 15 27 87595

M. Triggianese

Unit	Bouwkunde
Department	Building Knowledge
Telephone	+31 15 27 87468

C.M.H. Unal

Unit	Civiele Techniek & Geowetensch
Department	Atmospheric Remote Sensing
Telephone	+31 15 27 86257
Room	23.HG 2.18

A. Varveri

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 84665
Room	23.S2 1.26

Prof.dr. W.W. Veeneman

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 87754
Room	31.b2.110

J.R.T. van der Velde

Unit	Bouwkunde
Department	Landschapsarchitectuur
Telephone	+31 15 27 87902

Dr.ir. J.A.E. ten Veldhuis

Unit	Civiele Techniek & Geowetensch
Department	Water Systems Monitoring & Modelling
Telephone	+31 15 27 84734

Dr. M. Vizcaino

Unit	Civiele Techniek & Geowetensch
Department	Physical and Space Geodesy
Telephone	+31 15 27 85992
Room	23.HG 2.24

Dr. J.M. Vleugel

Unit	Civiele Techniek & Geowetensch
Department	Transport, Mobility and Logistics
Telephone	+31 15 27 86487
Room	23.HG 4.18

Dr.ing. M.Z. Voorendt

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 83340
Room	23.HG 3.76

Dr. H.G. van der Voort

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 88541
Room	31.b2.140

Dr. D. Wüthrich

Department	Hydraulic Structures and Flood Risk
-------------------	-------------------------------------

Dr. F. Zhang

Department	Concrete Structures
Department	Concrete Structures
Unit	Civiele Techniek & Geowetensch
Department	Concrete Structures
Room	23.S2 2.04

Dr.ir. H. Ziar

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 81907

